

FHV

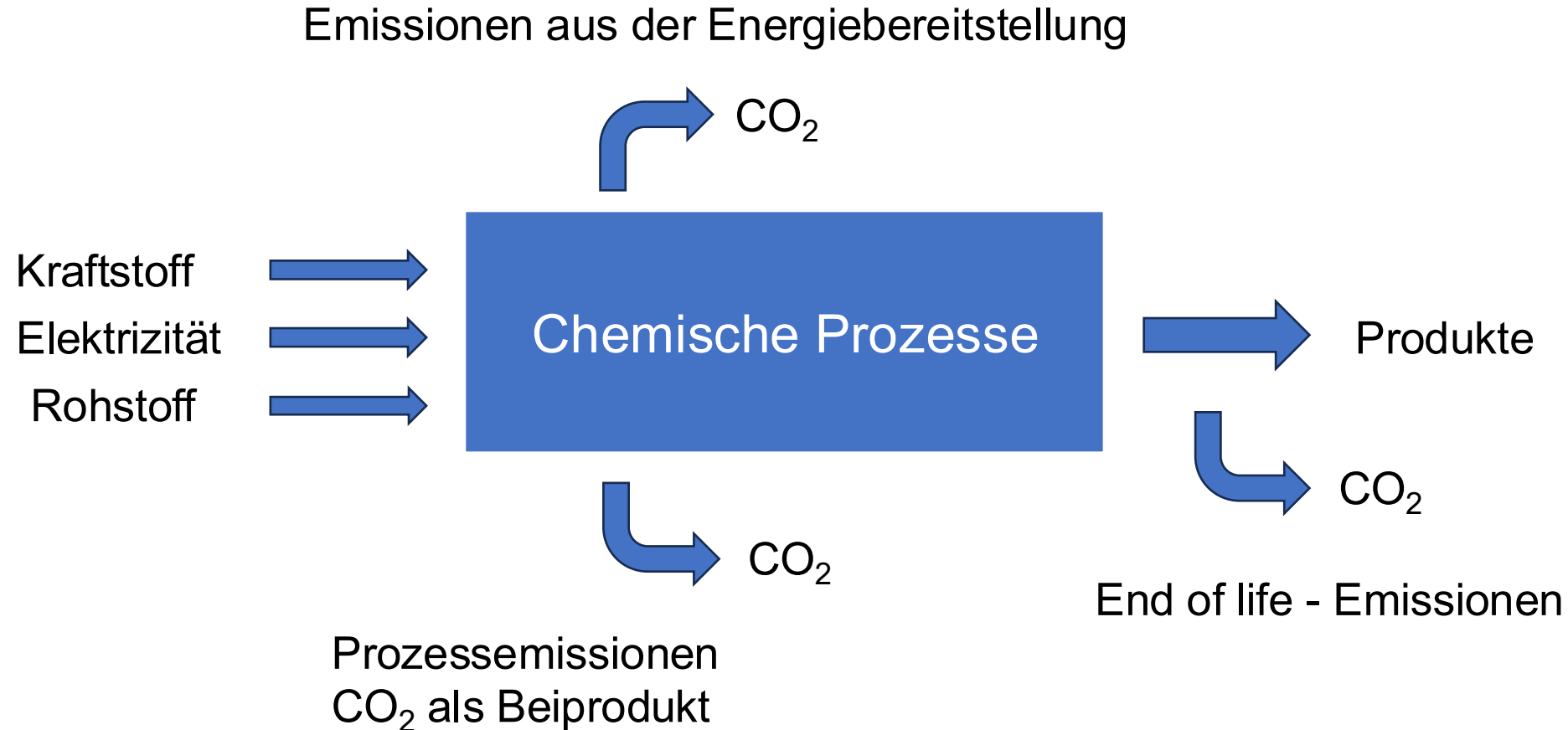
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Carbon Capture

Einleitung

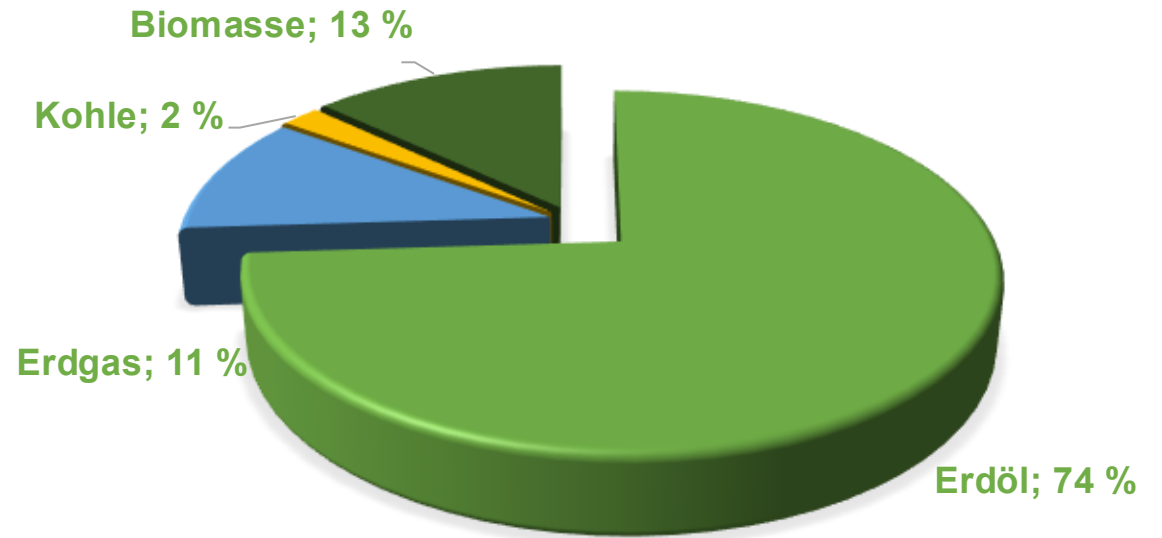
Industrie und CO₂-Emissionen



Chemische Industrie und CO₂-Emissionen

- Drittgrößter CO₂ Emitter (1,5 Gigatonnen pro Jahr)
- Erdölverbrauch: 14 % der Förderquote
- Erdgasverbrauch: 8 % der Förderquote
- 50 % als Rohstoff / 50 % als Energiequelle
- 85 % der Chemikalien aus Kohlenstoff
- Erdöl: Kunststoffe
- Erdgas: Methanol, Ammoniak
- Kohle: China

Deutsche Chemieindustrie



Kohlenstoffquelle



Biomasse



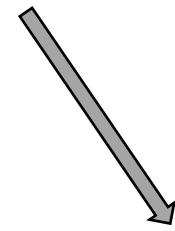
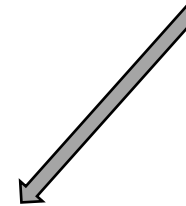
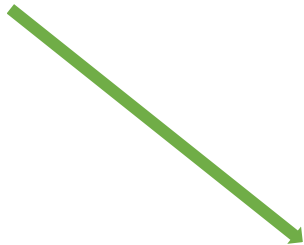
Recycling



Industrieabgase



Erneuerbare Energie



Chemikalien



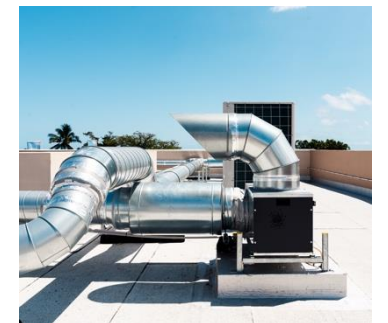
Treibstoffe



Chemikalien

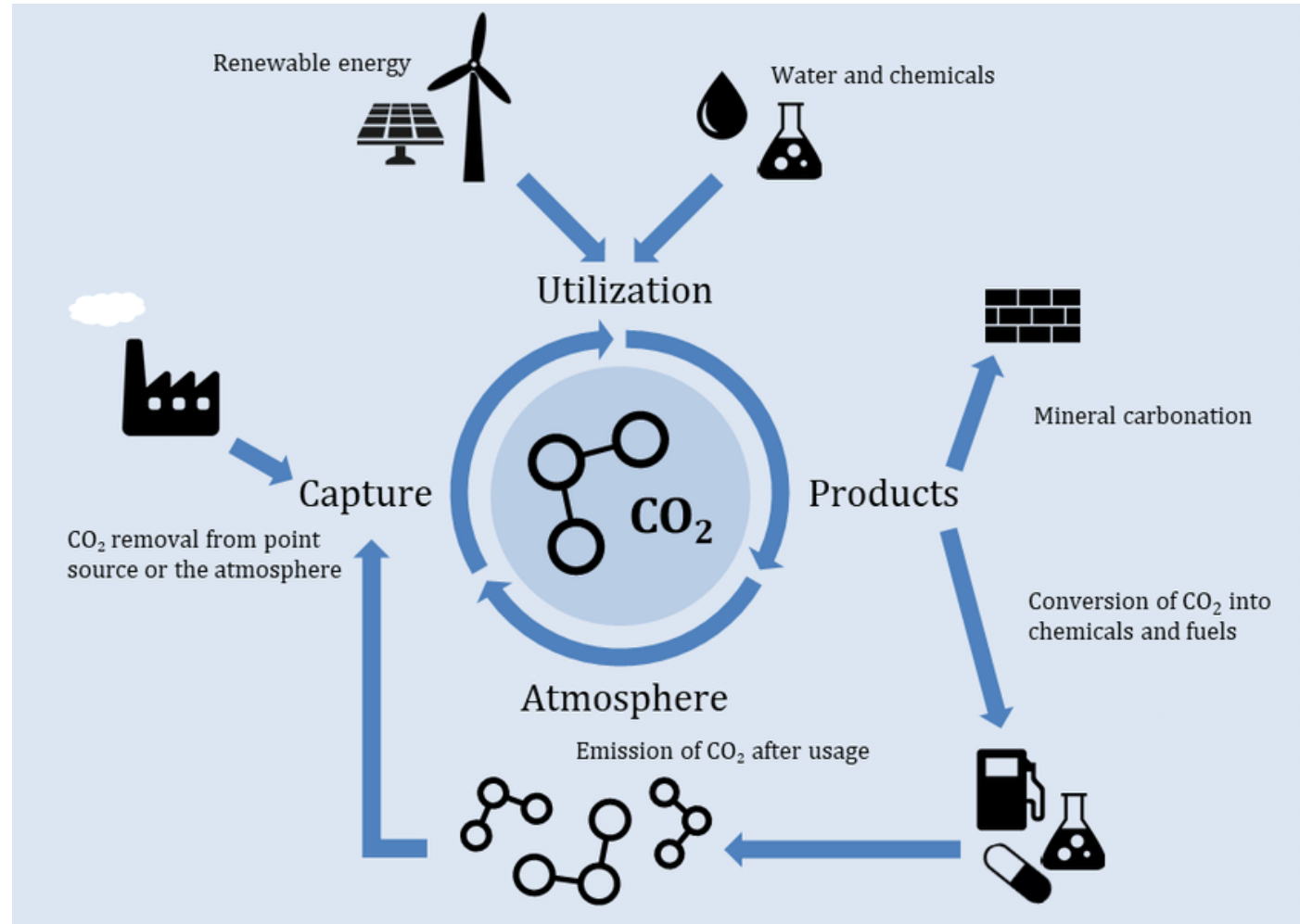


Treibstoffe

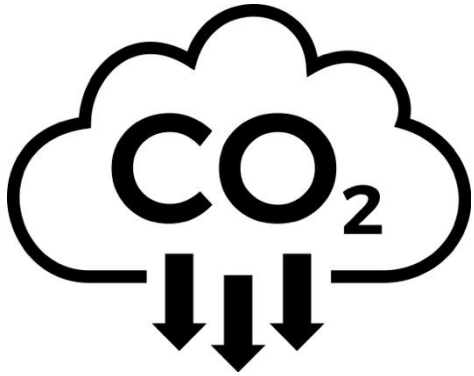


Power to Heat

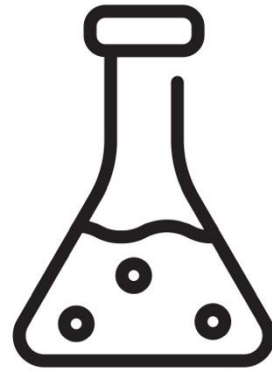
Carbon Capture and Utilisation/Storage



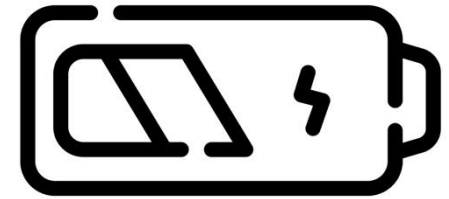
Motivation für die Nutzung von CO₂



- Reduktion der CO₂-Emissionen
- Ermöglichen einer „Low Carbon Economy“



- Ersetzen von fossilen Energiequellen
- Rohstoffquellen erweitern



- PtX für die Energiespeicherung
- Synergien schaffen

FHV

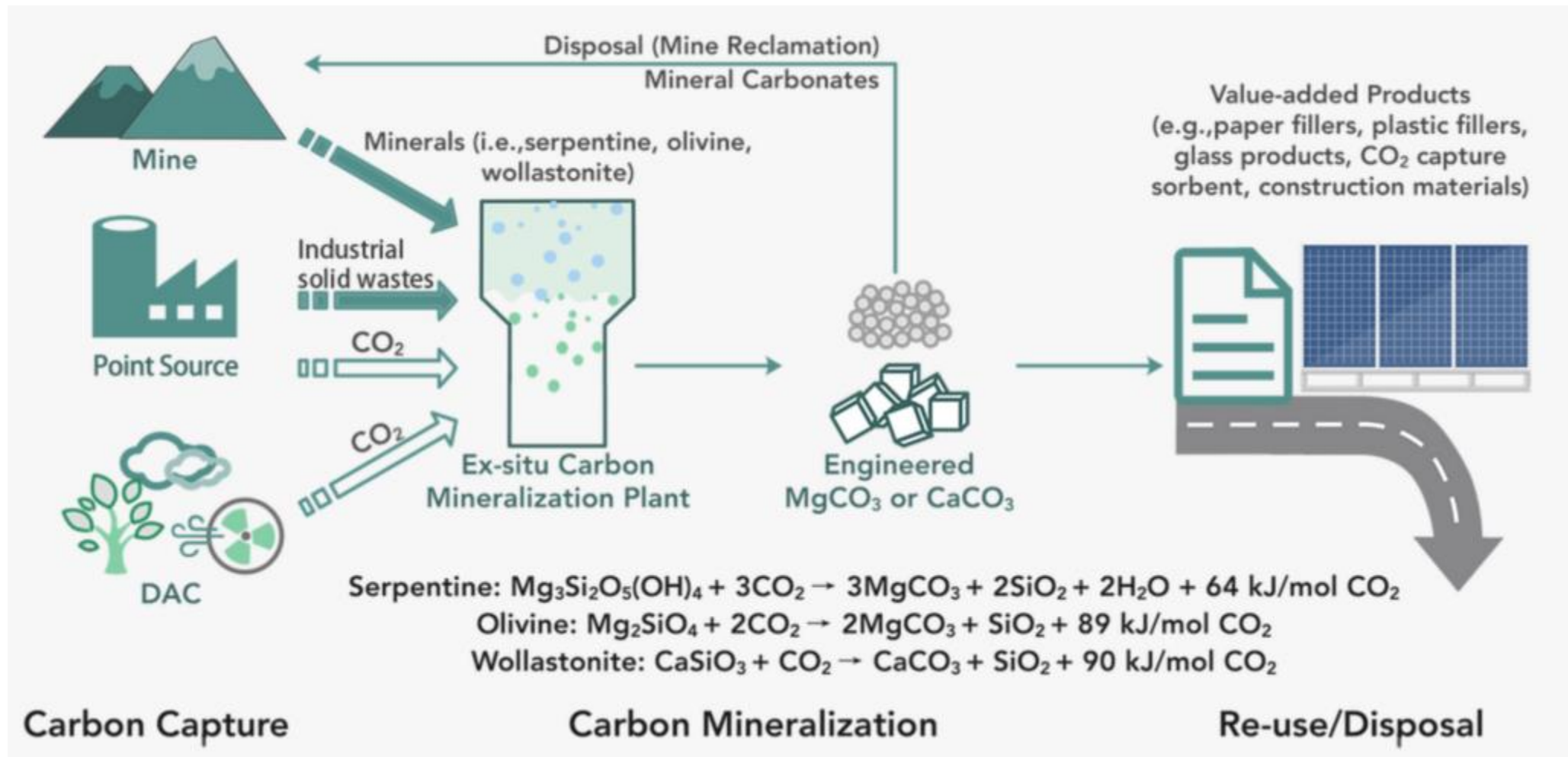
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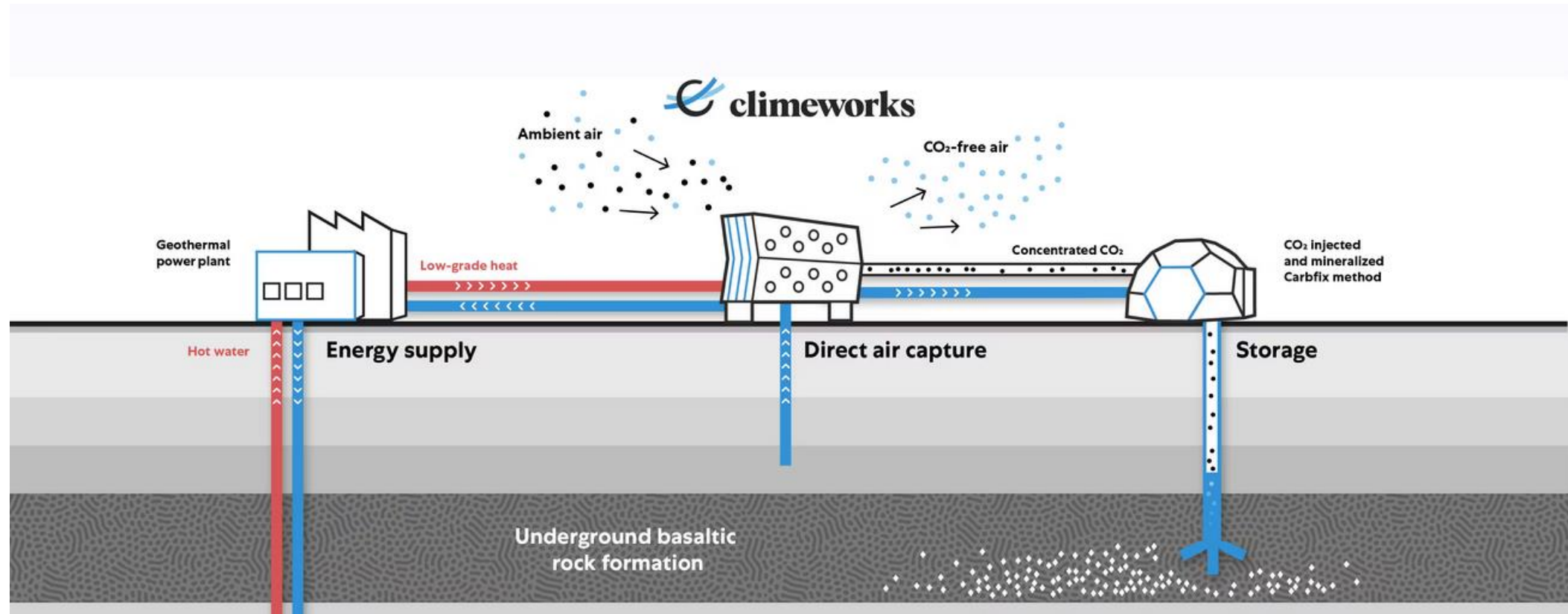
Carbon Capture

And Storage

Reaktionen mit CO₂ – Carbon Capture and Storage



Reaktionen mit CO₂ – Carbon Capture and Storage



Reaktionen mit CO₂ – Carbon Capture and Storage

CarbFix (Island)

Abscheidung

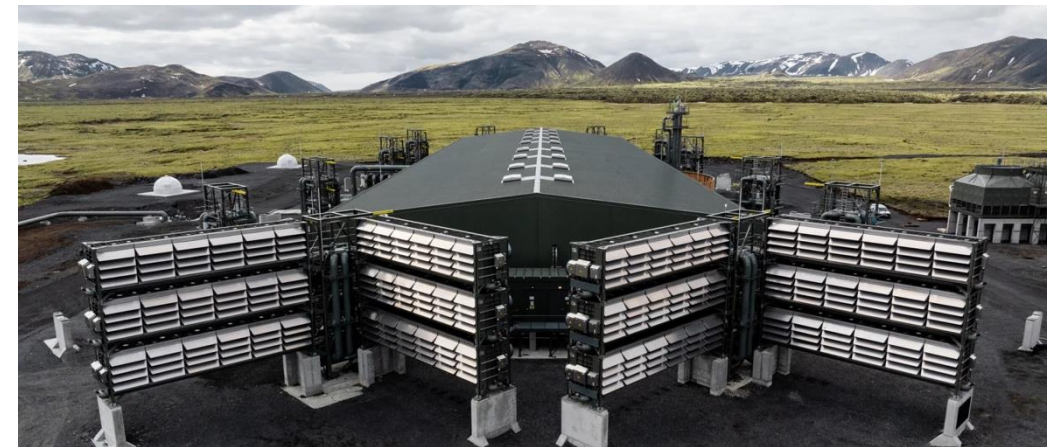
- Geothermiekraftwerk durch Druckentlastung (10-20 €/t)
- DAC – Aminwäsche (40 bis 80 €/t)

Speicherung

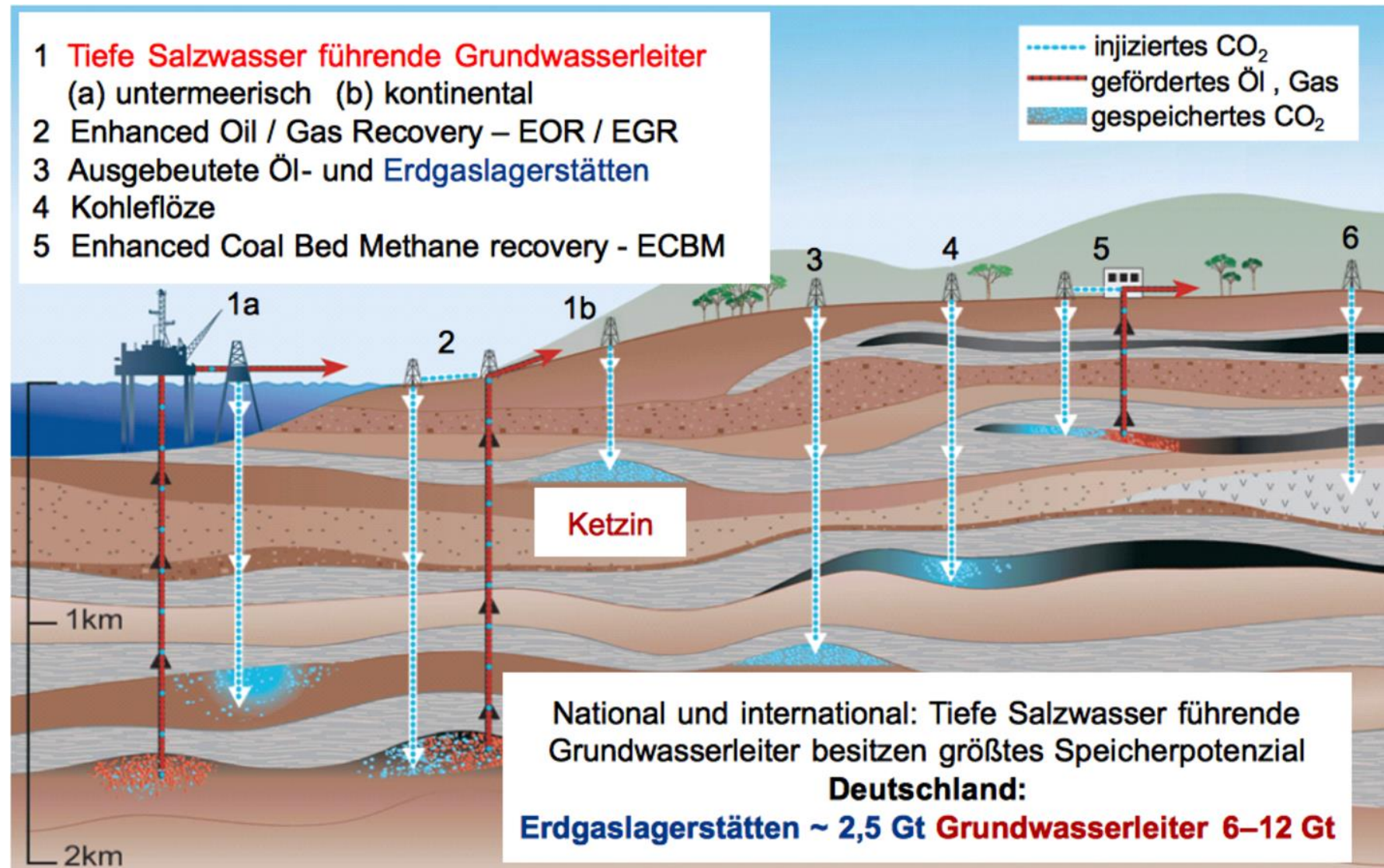
- Mineralisierung im Basaltgestein (20 bis 25 €/t)

Abscheidepotential

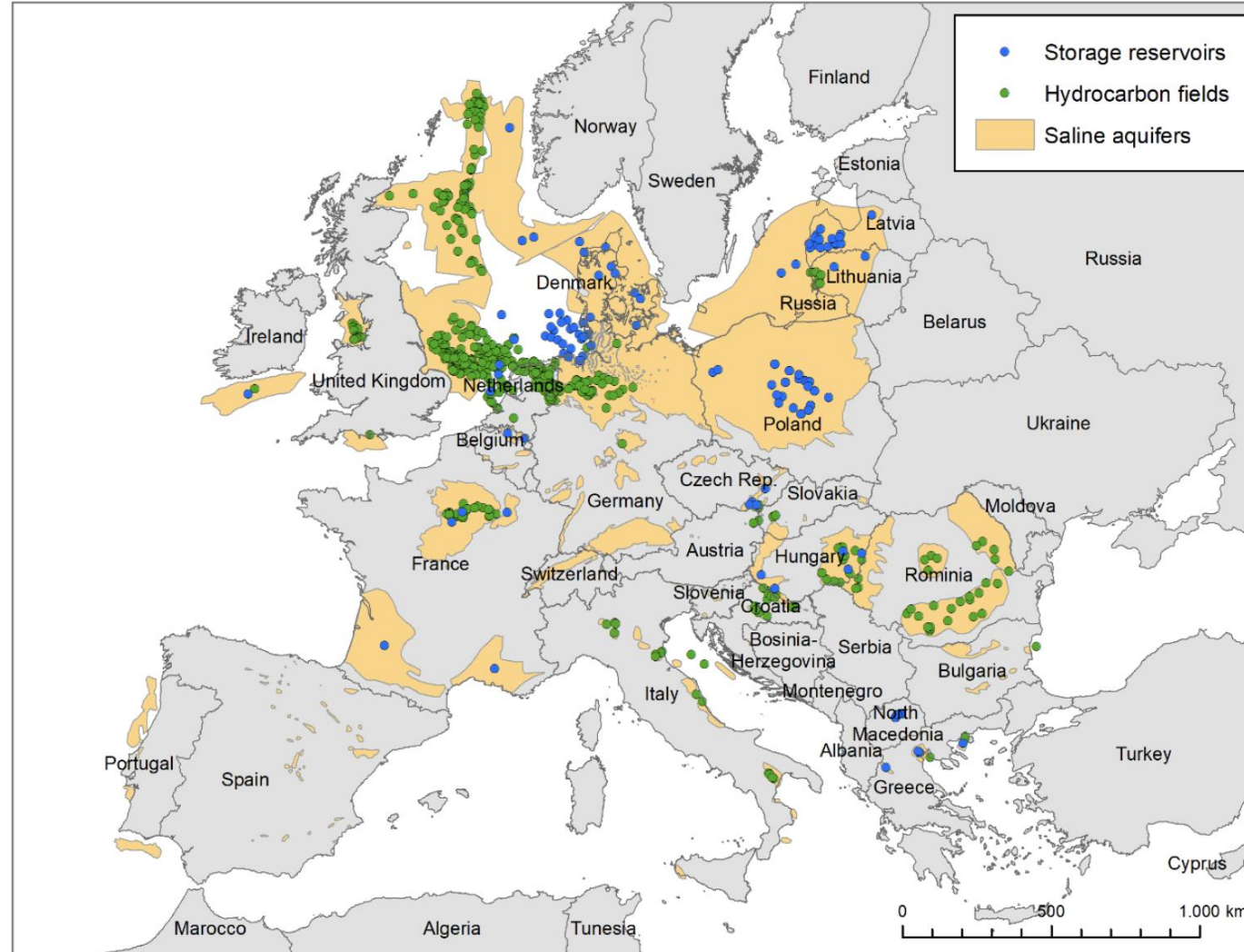
- 36.000 t CO₂/Jahr



Reaktionen mit CO₂ – Carbon Capture and Storage



Reaktionen mit CO₂ – Carbon Capture and Storage



Reaktionen mit CO₂ – Carbon Capture and Storage

Sleipner (Norwegen)

Abscheidung

- Aus dem eigenen geförderten Erdgas durch Aminwäsche

Speicherung

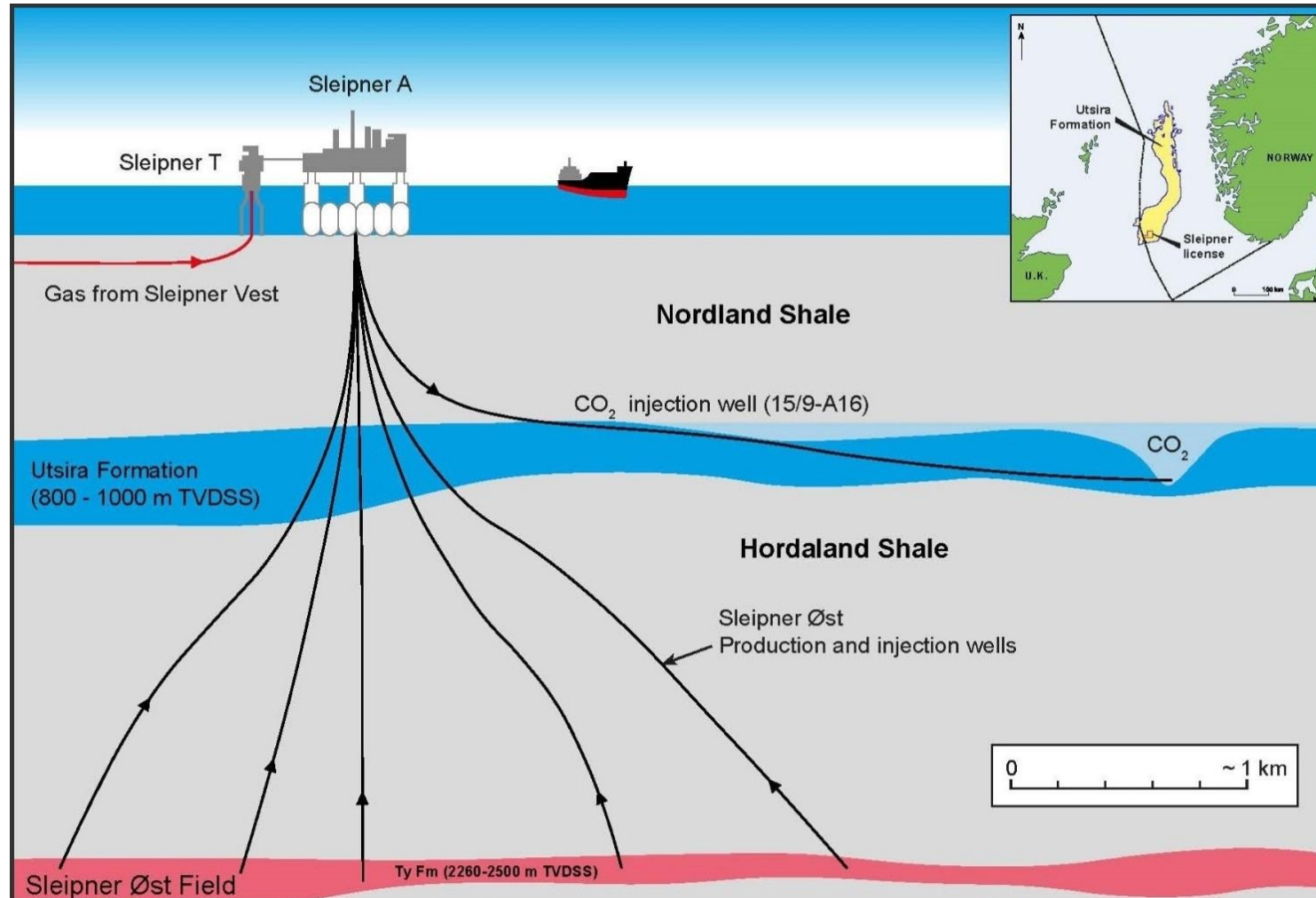
- Geologisch in salinen Aquiferen unter dem Meeresboden in ca. 1000 m Tiefe

Zahlen: sehr schwierige Grenzziehung ca. 50 €/tonne

Abscheidepotential

- 900.000 t CO₂/Jahr (15,5 Mt bisher)

Reaktionen mit CO₂ – Carbon Capture and Storage



Reaktionen mit CO₂ – Carbon Capture and Storage

Weyburn-Midale CO₂ EOR Project (Kanada)

Abscheidung & Transport

- Solexol Verfahren + 300 km Pipeline

Speicherung

- Enhanced Oil Recovery

Zahlen: sehr geringe Kosten durch sehr günstige Umstände

Abscheidepotential

- 2 - 3 Mt CO₂/Jahr (30 Mt bisher)

Reaktionen mit CO₂ – Carbon Capture and Storage



FHV

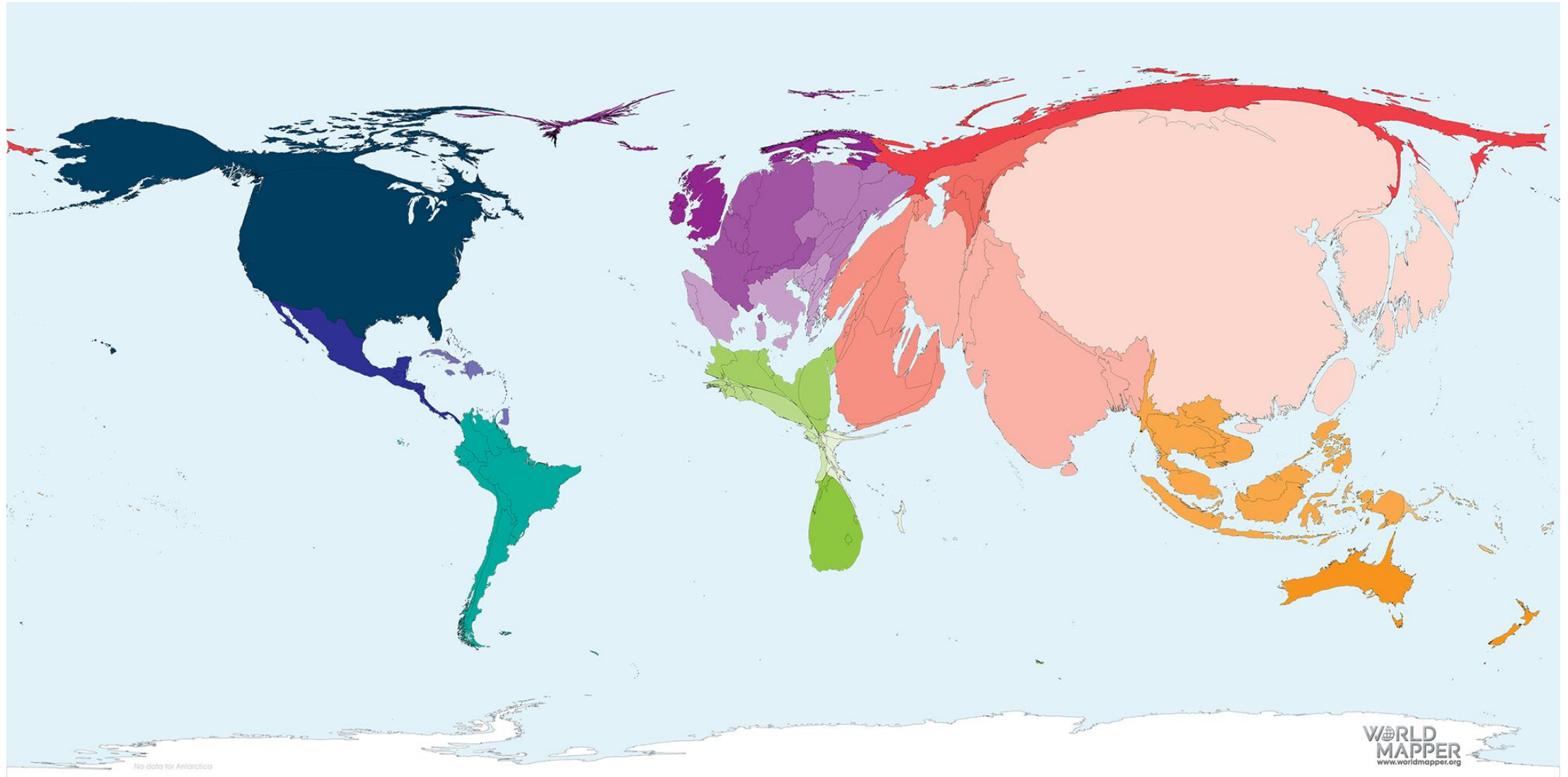
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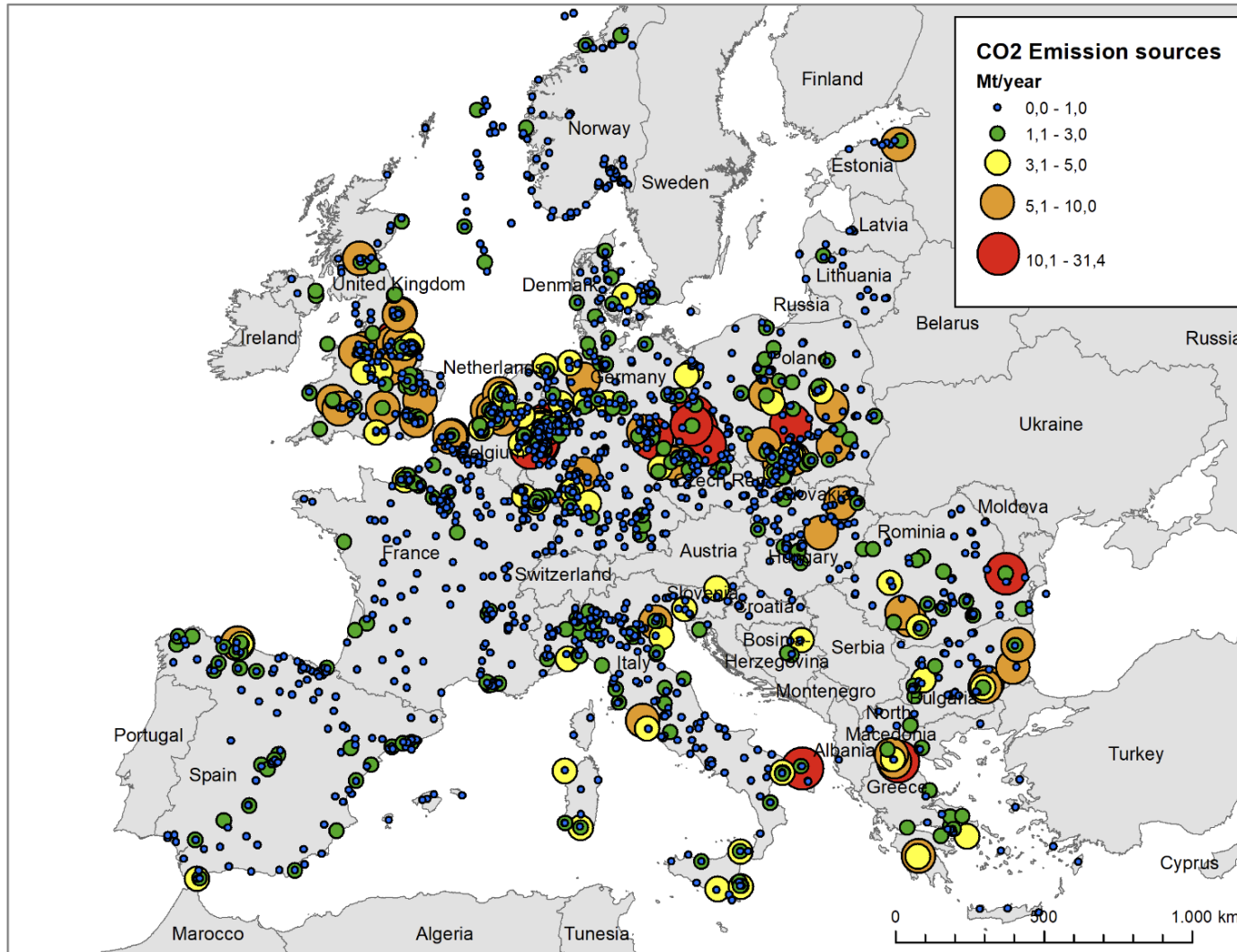
Carbon Capture

Prozesse

CO₂-Emissionen - Weltweit



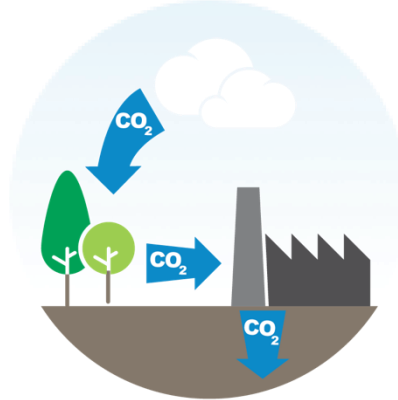
CO₂-Emissionen - Europa



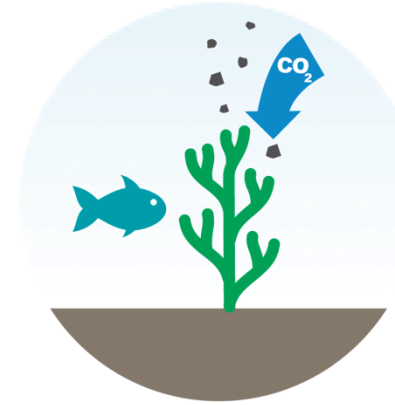
CO₂-Emissionen - Europa



Afforestation and Reforestation
Tree growth takes up CO₂ from the atmosphere.



Bioenergy with carbon capture and storage
Plants turn CO₂ into biomass that fuel power plants. CO₂ captured and stored underground.



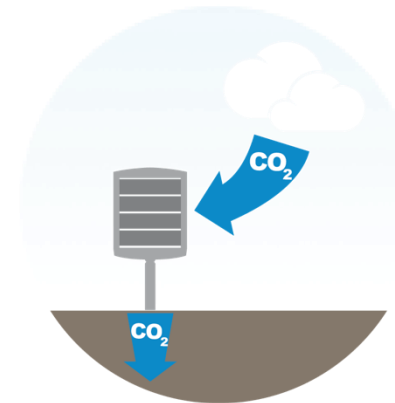
Ocean fertilization
Iron or other nutrients are applied to the ocean increasing CO₂ absorption.



Biochar
Partly burnt biomass is added to soils absorbing additional CO₂.

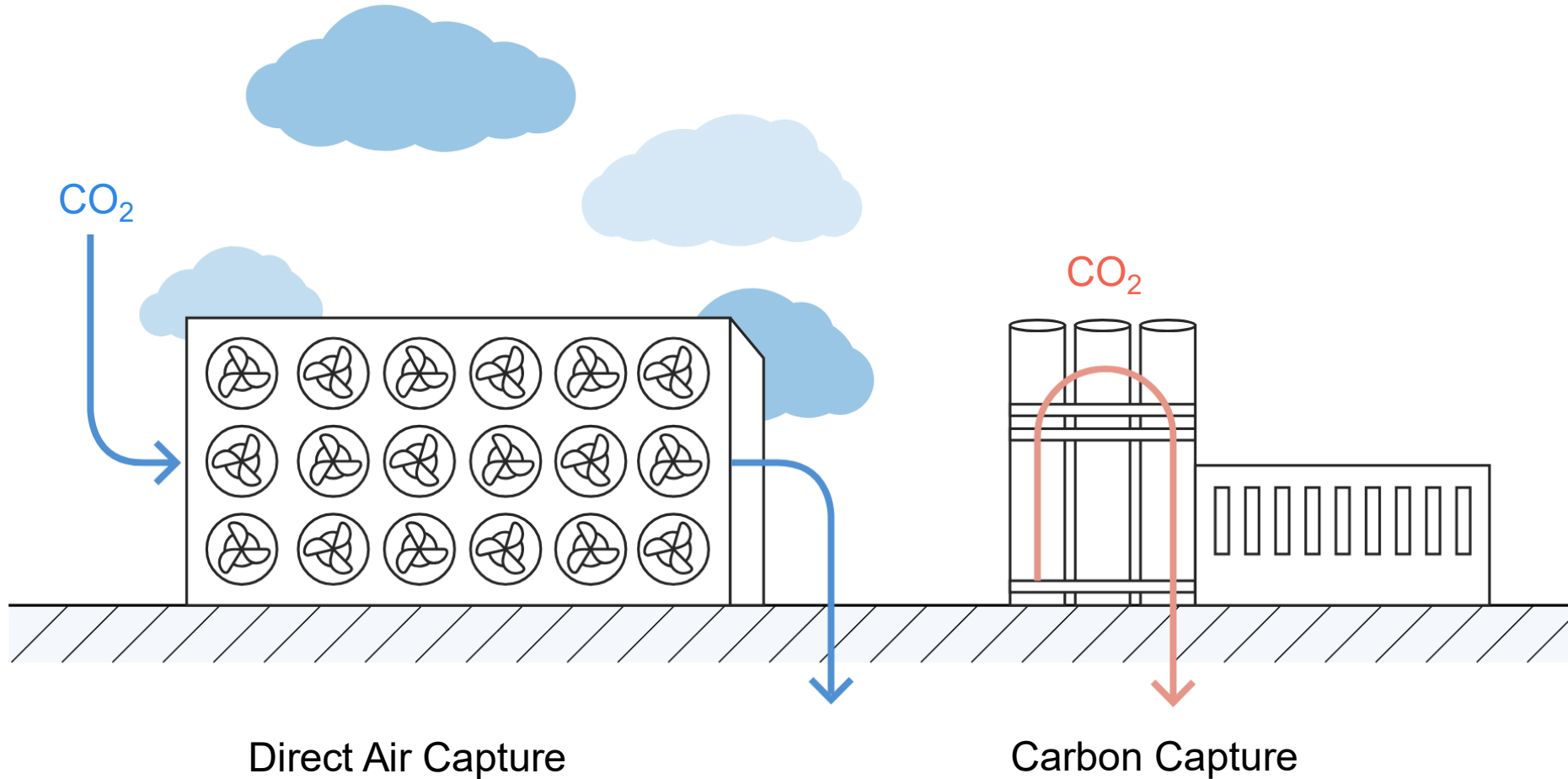


Enhanced weathering
Crushed minerals are applied to soil for chemical CO₂ absorption.



Direct Air Capture
CO₂ is removed from ambient air through chemical processes and stored underground.

CO₂-Emissionen - Europa



CO₂-Emissionen - Europa

	Area required to remove 8 Gt CO ₂ per year	Water required to remove 8 Gt CO ₂ per year	Expected cost at large scale	Impact on environment*
 Afforestation Large-scale tree plantations to increase carbon storage in biomass and soil.	 6,400,000 km² Europe = 10,500,000 km ²	 740 km³ Yearly global freshwater withdrawal 2010 = 4,000 km ³	 5–50 USD/t CO₂	 Biodiversity  Albedo  Food Security
 BECCS Bioenergy in combination with Carbon Capture and Storage.	 2,500,000 km²	 480 km³	 100–200 USD/t CO₂	 Biodiversity  Albedo  Food security
 Enhanced weathering Distribution of crushed silicate rocks on soil surfaces to absorb and bind CO ₂ chemically.	 220,000 km²	 3 km³	 50–200 USD/t CO₂	 River/ocean chemistry
 Direct air capture Direct capture of CO ₂ from ambient air through engineered chemical reactions.	 15,800km^{2**}	 8 km³ Potentially zero	 < 200 USD/t CO₂	 Minor

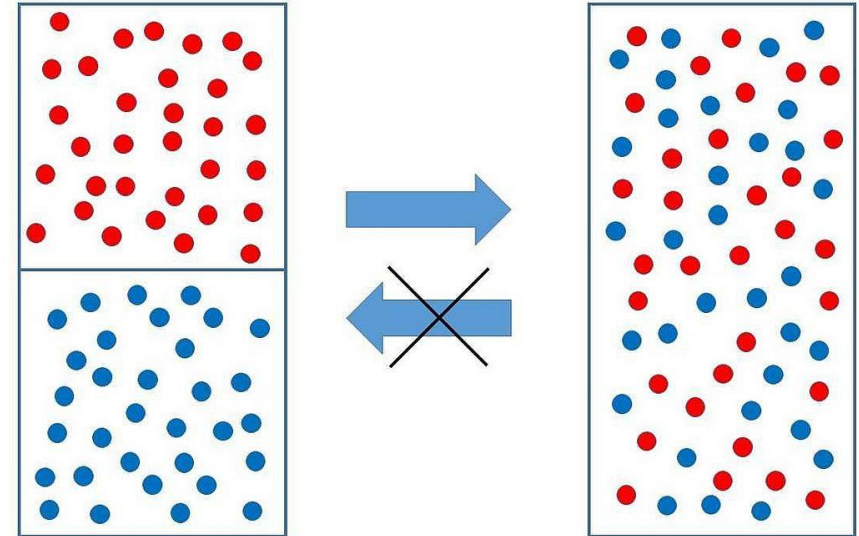


Entropie

Wie viel Energie brauchen wir denn?

CO₂ – Konzentration:

- Biogas: ca. 40 %
- Verbrennung: ca. 15 %
- Oxy-Combustion Verbrennung: ca. 70 %
- Luft: 0,04 %



[Video – 7:12](#)

Wie viel Energie brauchen wir denn?

CO₂ – Konzentration:

- Biogas: ca. 40 %
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 - Oxy-Combustion Verbrennung: ca. 70 %
 - Luft: 0,04 %
-
- Wieviel Energie benötigen wir (mindestens) um 1 Tonne CO₂ aufzufangen

Wie viel Energie brauchen wir denn?

$$w_{\min} = RT \ln \left(\frac{1}{y_{CO_2}} \right)$$

$$RT \approx 8,314 \cdot 298 = 2478 \text{ J/mol}$$

$$n = \frac{1,000,000 \text{ g}}{44,01 \text{ g/mol}} \approx 22,722 \text{ mol}$$

Wie viel Energie brauchen wir denn?

$$y_{CO_2} = 0,40$$

$$w_{\min} = RT \ln(1/0,40) = 2478 \cdot 0,916 \approx 2270 \text{ J/mol}$$

Für 1 Tonne CO₂:

$$2270 \cdot 22722 \approx 5,16 \cdot 10^7 \text{ J}$$

$$\approx 51,6 \text{ MJ}$$

$$\approx 14,3 \text{ kWh/t}_{CO_2}$$

$$y_{CO_2} = 0,15$$

$$\approx 29,7 \text{ kWh/t}_{CO_2}$$

$$y_{CO_2} = 0,70$$

$$\approx 5,6 \text{ kWh/t}_{CO_2}$$

$$y_{CO_2} = 0,0004$$

$$\approx 122 \text{ kWh/t}_{CO_2}$$

FHV

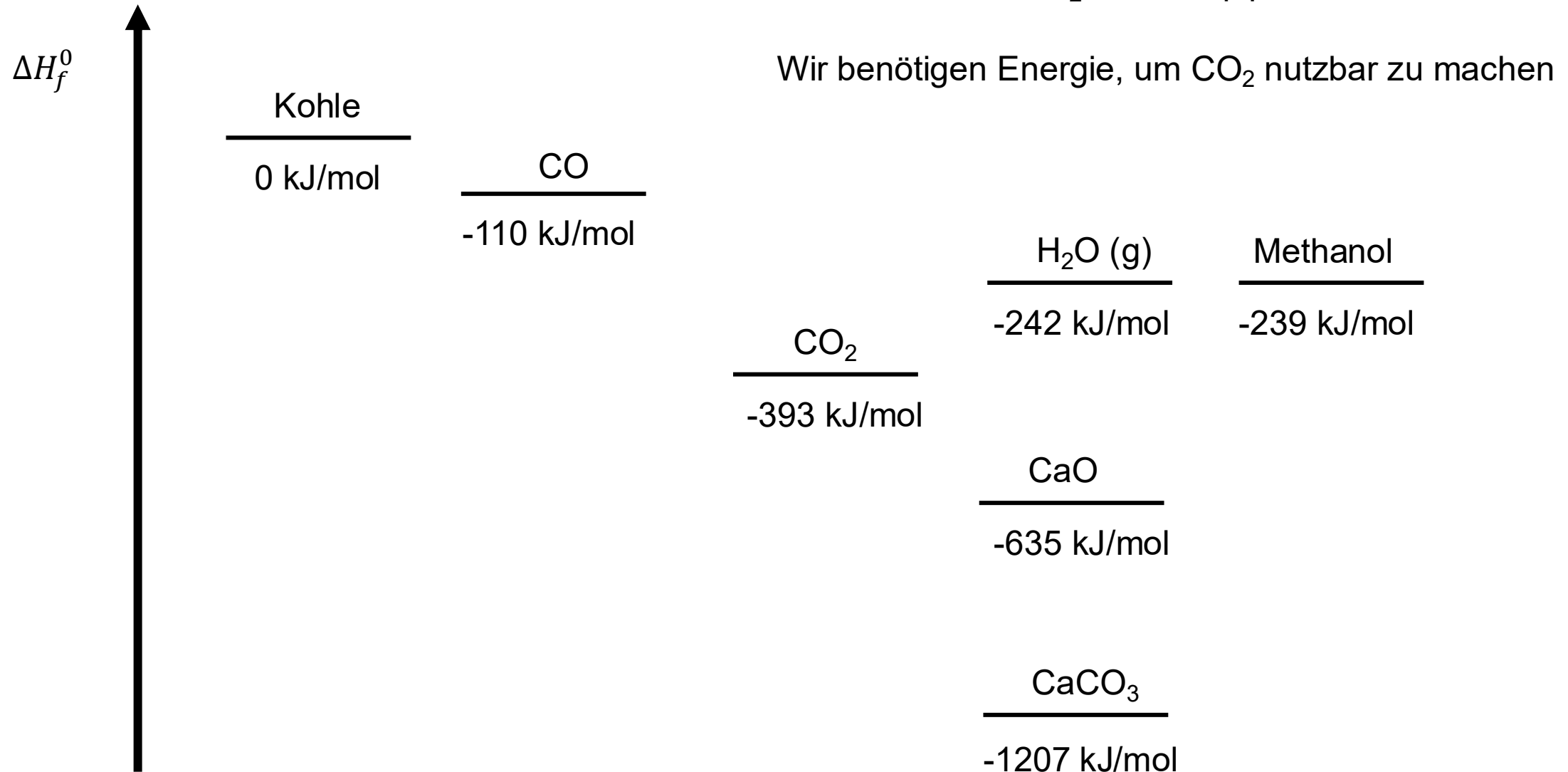
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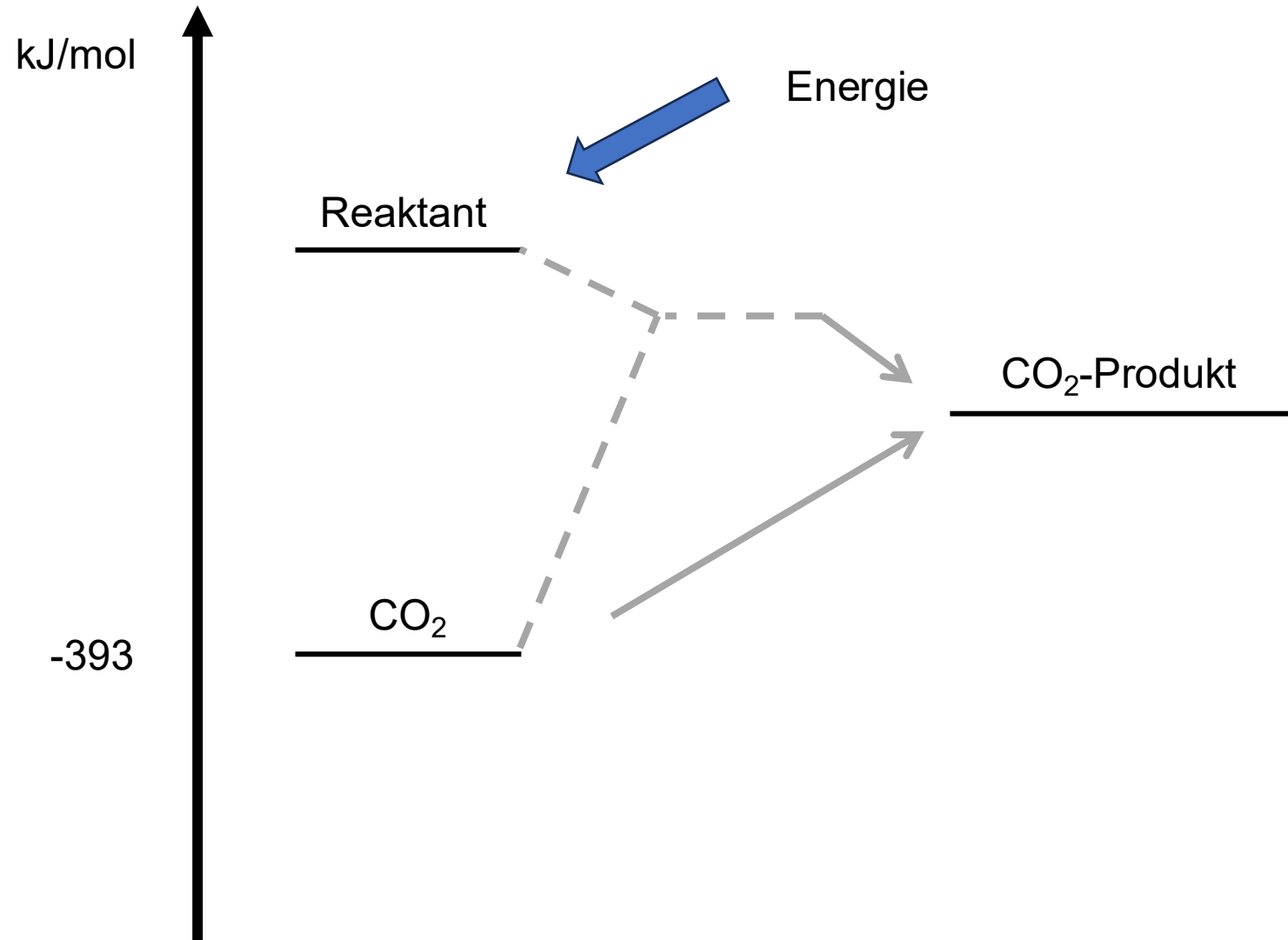
Carbon Capture

Thermodynamische
Aspekte

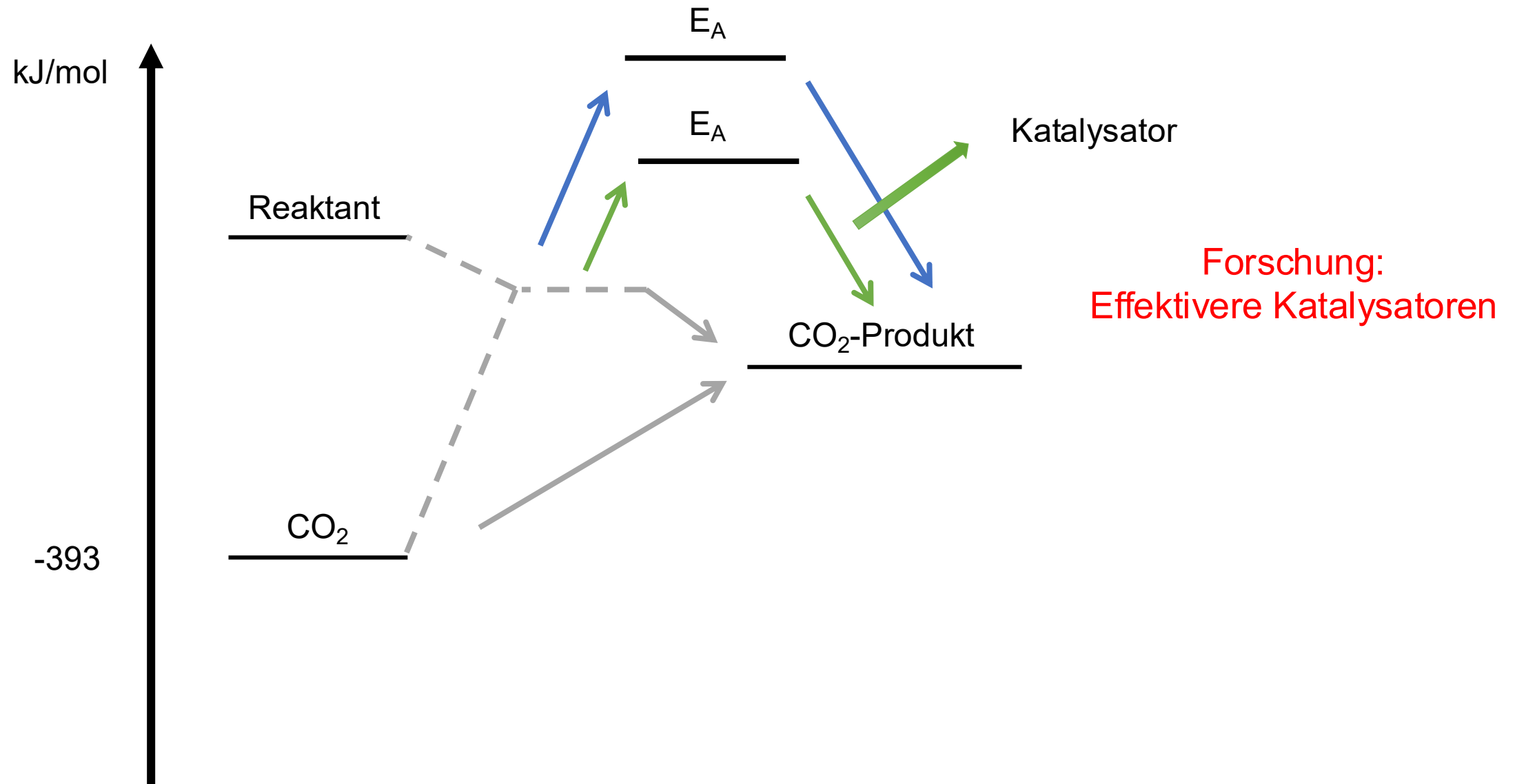
Thermodynamik



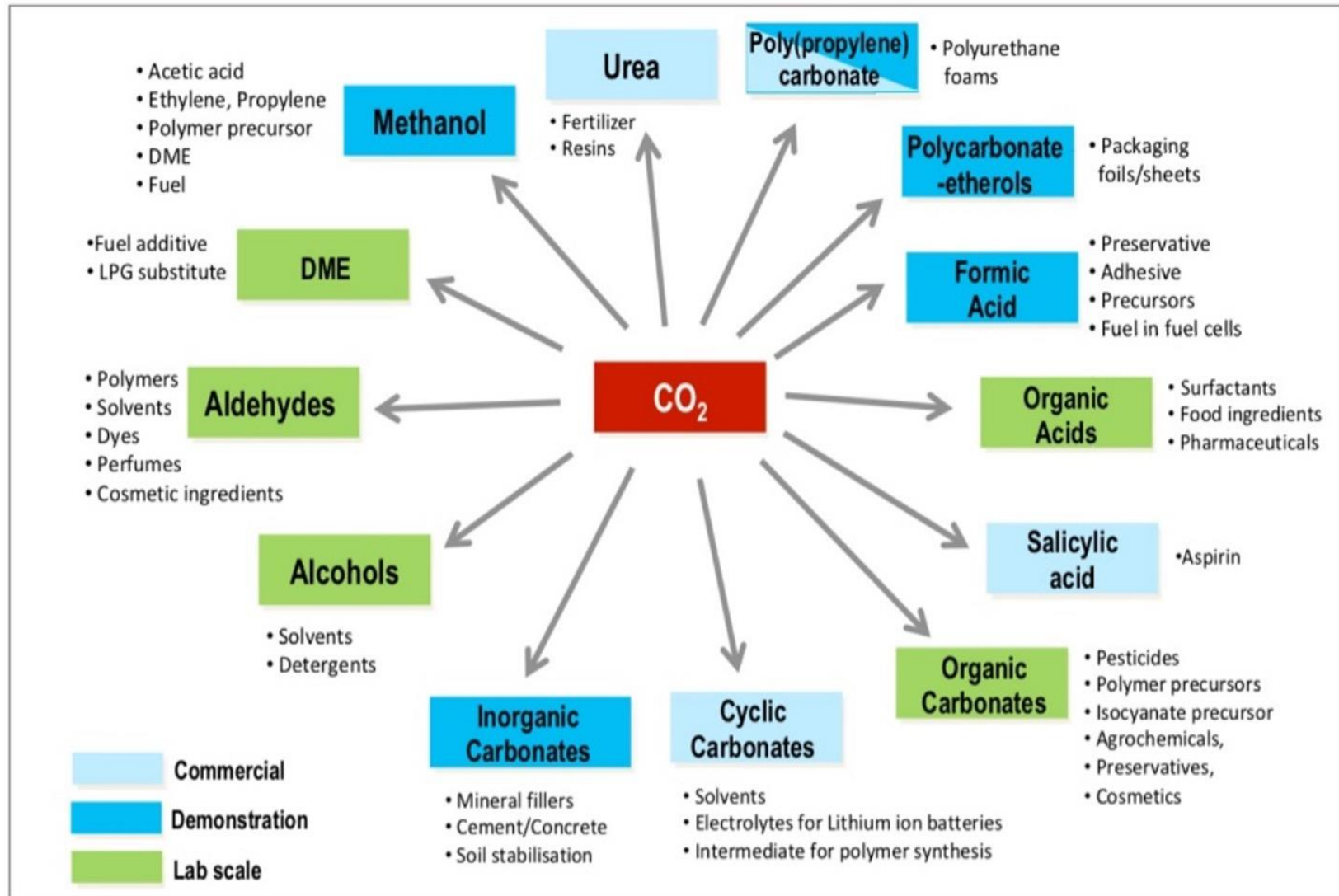
Thermodynamik



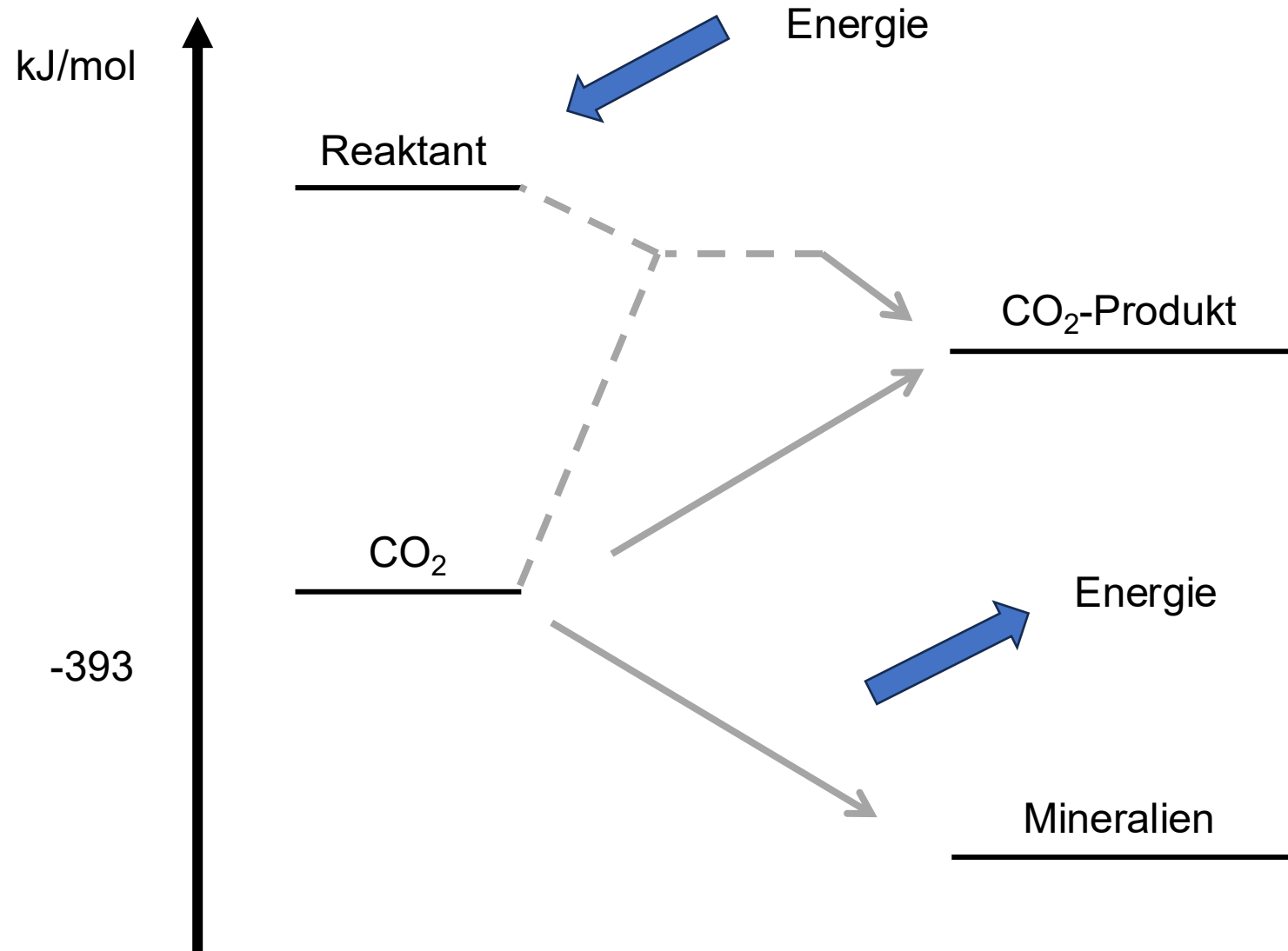
Thermodynamik



Reaktionen mit CO₂



Thermodynamik



Kinetisch gehindert
Benötigt Jahre / Jahrtausende

Bedingungen:
350 °C / 5 bar

Thermodynamik

Wassergas Shift Reaktion:

$$\Delta H_R^0 = 42,1 \text{ kJ/mol}$$



$$\Delta S_{\text{CO}_2}^0 = 212 \text{ J/mol} \quad / \quad \Delta S_{\text{H}_2}^0 = 131 \text{ J/mol} \quad / \quad \Delta S_{\text{CO}}^0 = 198 \text{ J/mol} \quad / \quad \Delta S_{\text{H}_2\text{O}}^0 = 189 \text{ J/mol}$$

$$\Delta S_R = 44 \text{ J/mol}$$

$$\Delta G_R = \Delta H_R - T * \Delta S \quad \longrightarrow \quad T = -\frac{\Delta H_R}{\Delta S}$$

$$T = -\frac{42100 \frac{\text{J}}{\text{mol}}}{44 \frac{\text{J}}{\text{mol} \cdot \text{K}}} = 957 \text{ K}$$



Power to X

Wasserstoff

Was ist Power to X

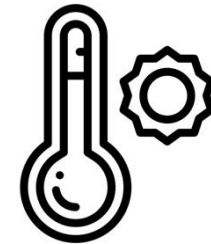
Power to X



X = Chemikalien

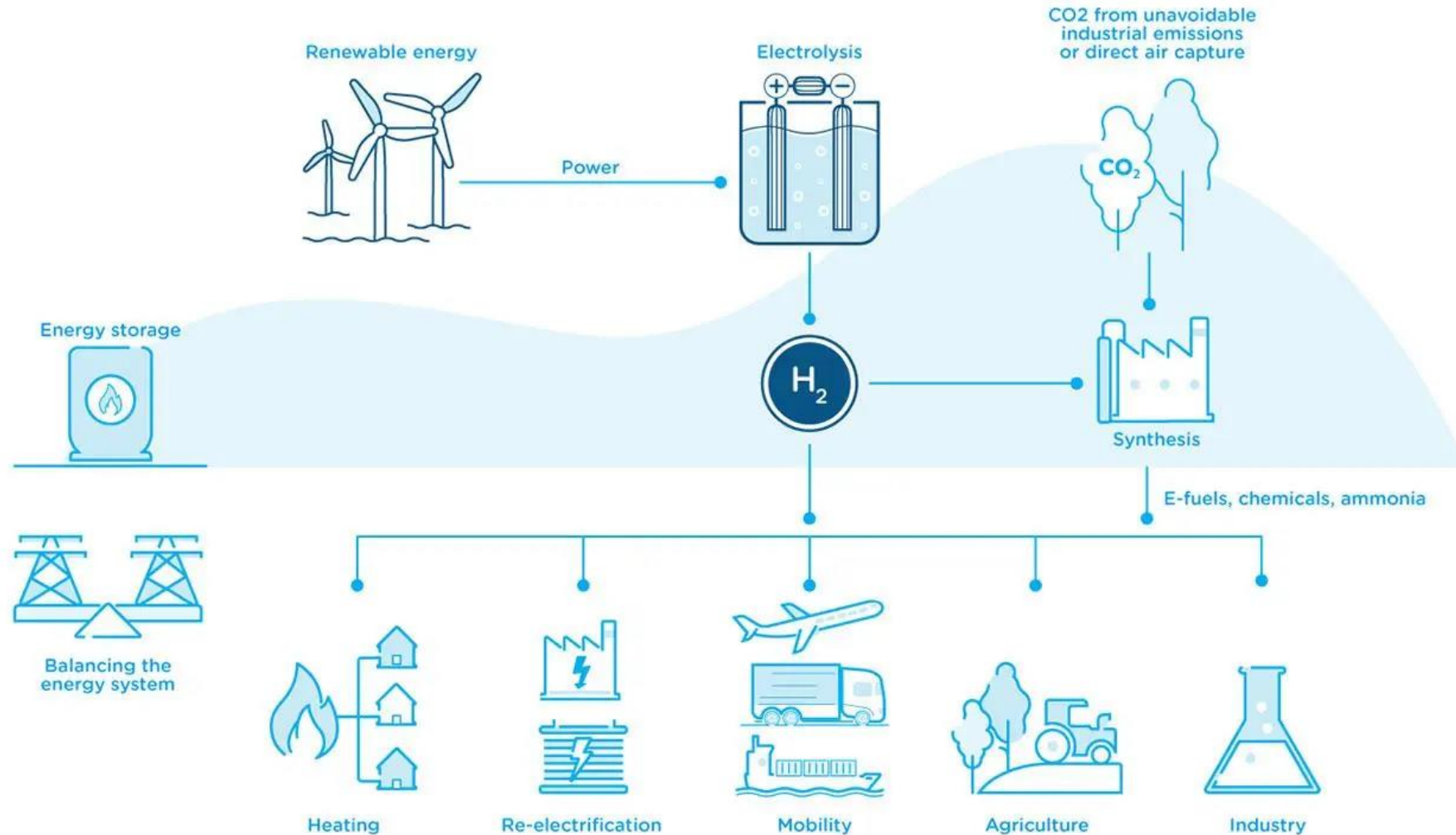


X = Gas (Methan / H₂)

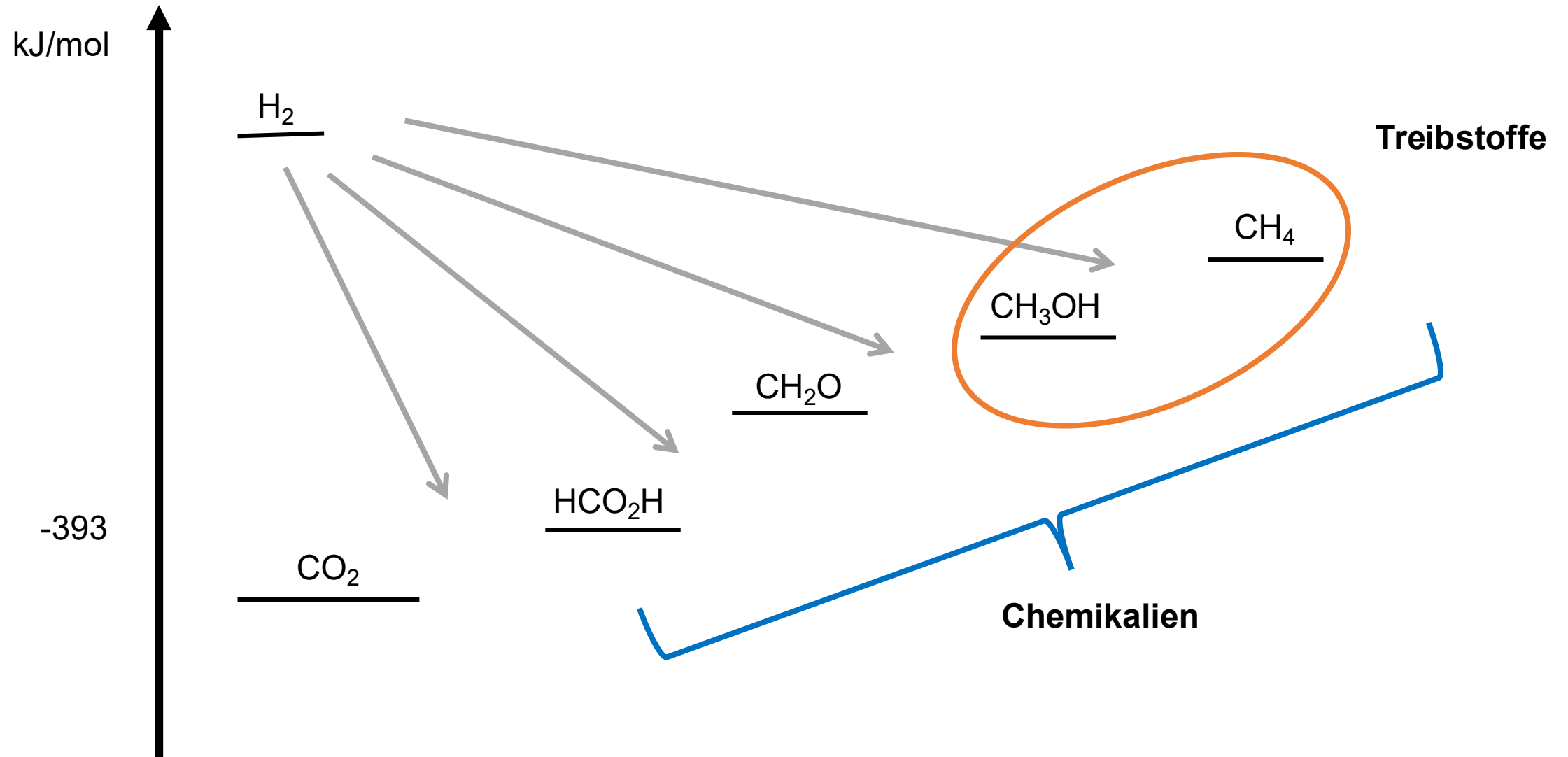


X = Wärme

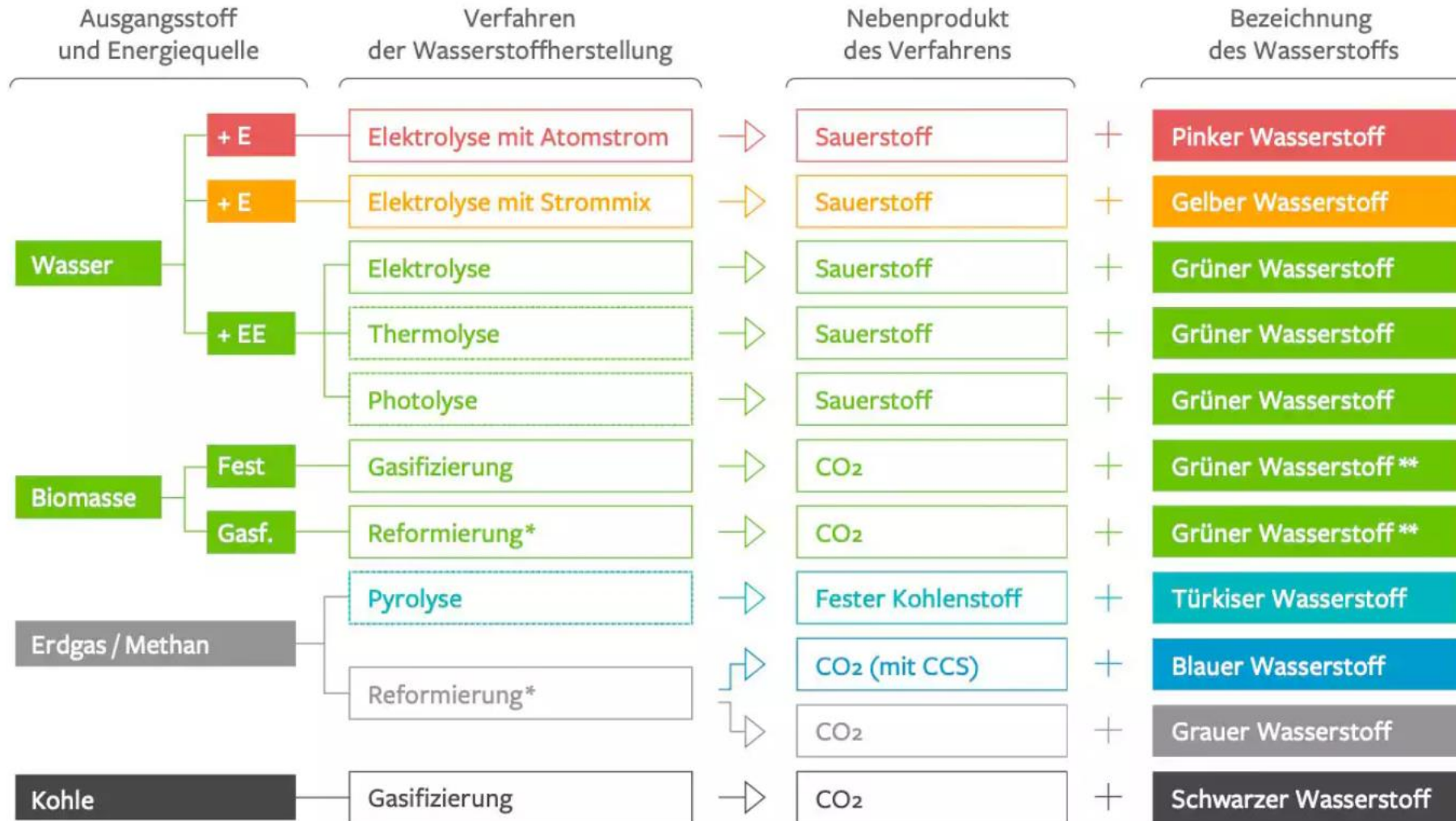
Was ist Power to X



Thermodynamik

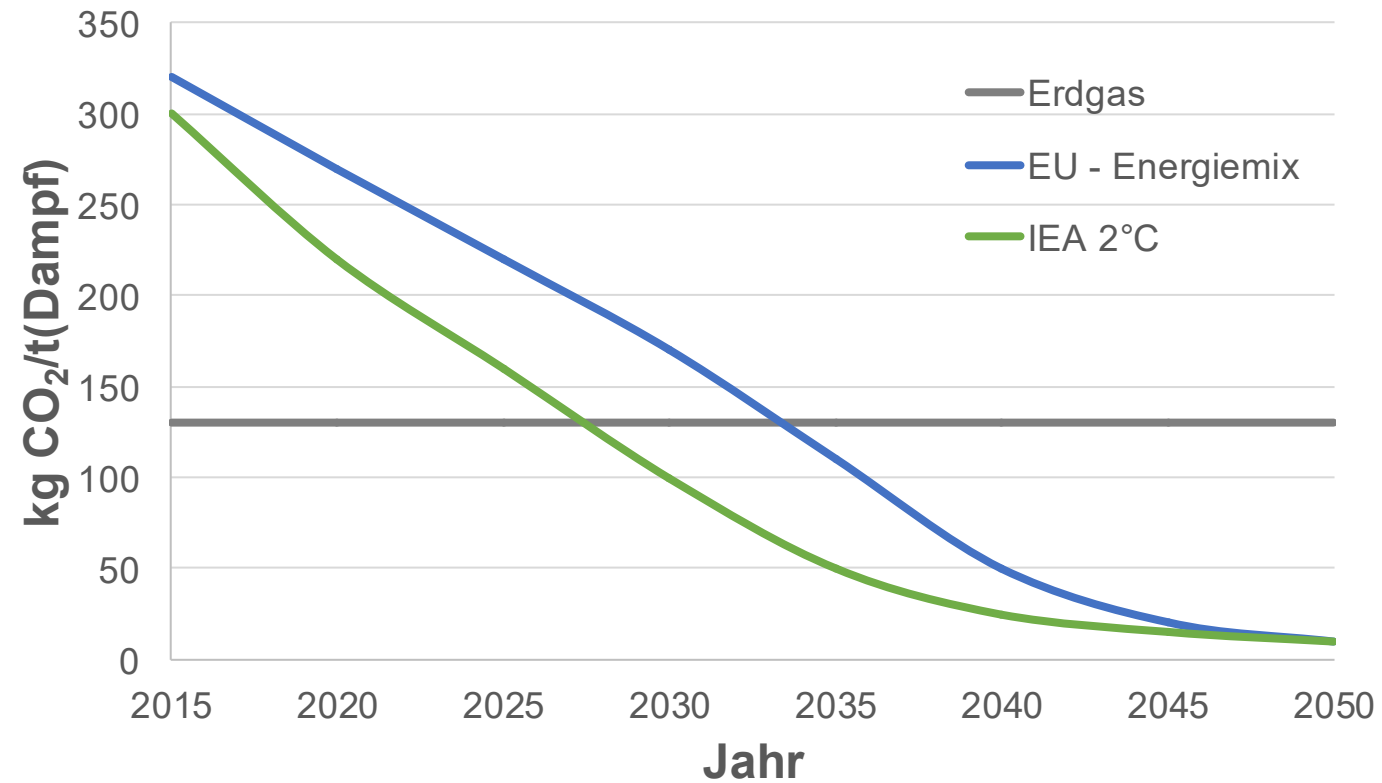


Wasserstoff - Farbenschema



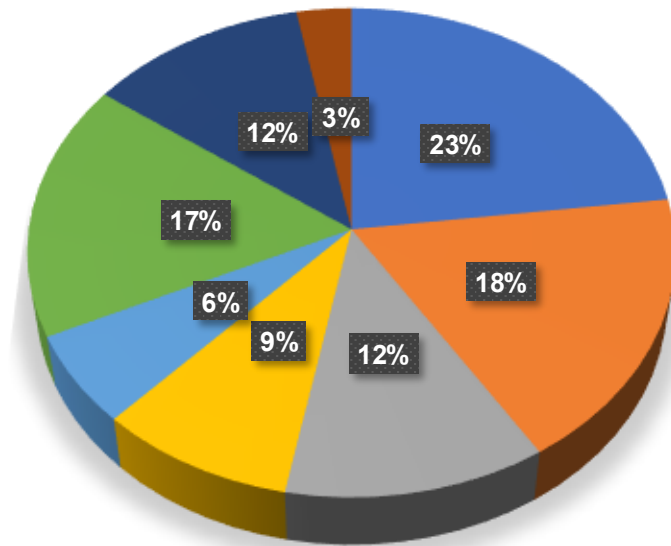
Erneuerbare Energiequellen

Power to Heat: für Illustrationszwecke



Strommix

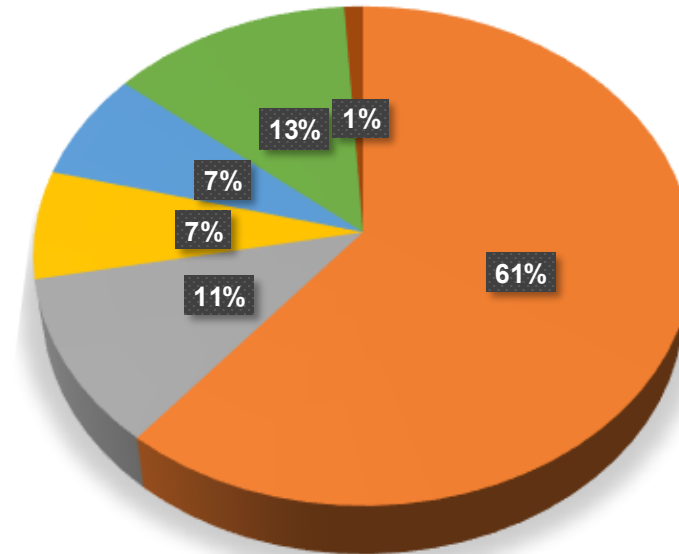
Europa



Kernenergie	Wasserkraft
Wind	Solar
Biomasse	Erdgas
Kohle	Erdöl

226 gCO₂/tWh

Österreich



Kernenergie	Wasserkraft
Wind	Solar
Biomasse	Erdgas
Kohle	Erdöl

76 gCO₂/tWh

FHV

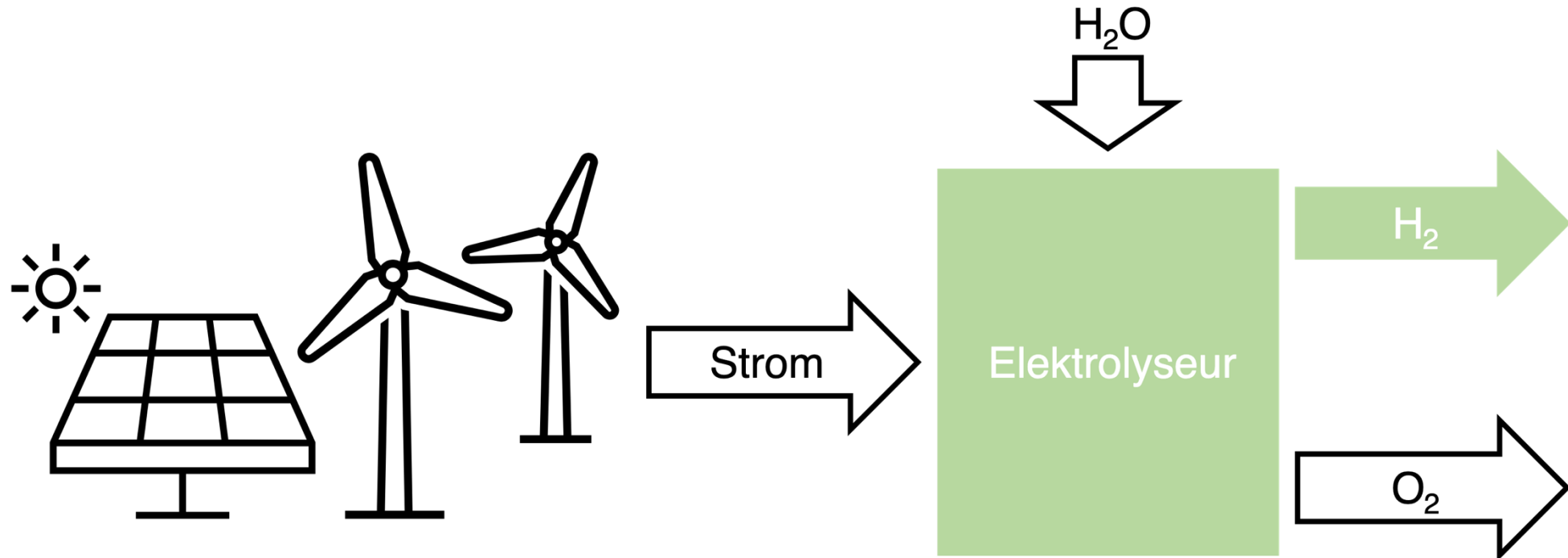
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Wasserstoff

Hydrolyse

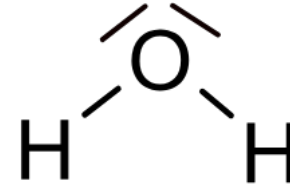
Grundlegendes zur Elektrolyse



Grundlegendes zur Elektrolyse

Zersetzungsspannung: 1,23 eV

- Kinetische Aktivierungsenergie
- Widerstand an der Elektrode
- Konzentration der Ionen



$$\Delta G_r = -n * F * \Delta E$$

Grundlegendes zur Elektrolyse

1. Energie pro Mol Wasserstoff

$$\Delta G = 2 \cdot 96485 \cdot 1,23$$

$$\Delta G \approx 237000 \text{ J/mol}$$

$$= 237 \text{ kJ/mol H}_2$$

2. Stoffmenge von 1 Tonne Wasserstoff

Molare Masse:

$$M(\text{H}_2) = 2 \text{ g/mol}$$

$$n = \frac{1,000,000 \text{ g}}{2 \text{ g/mol}} = 500,000 \text{ mol}$$

3. Gesamtenergie

$$E = 500,000 \cdot 237 \text{ kJ}$$

$$E = 118,500,000 \text{ kJ}$$

$$E = 118,500 \text{ MJ}$$

$$E = 118,5 \text{ GJ}$$

4. Umrechnung in kWh

$$1 \text{ kWh} = 3,6 \text{ MJ}$$

$$E = \frac{118,500}{3,6}$$

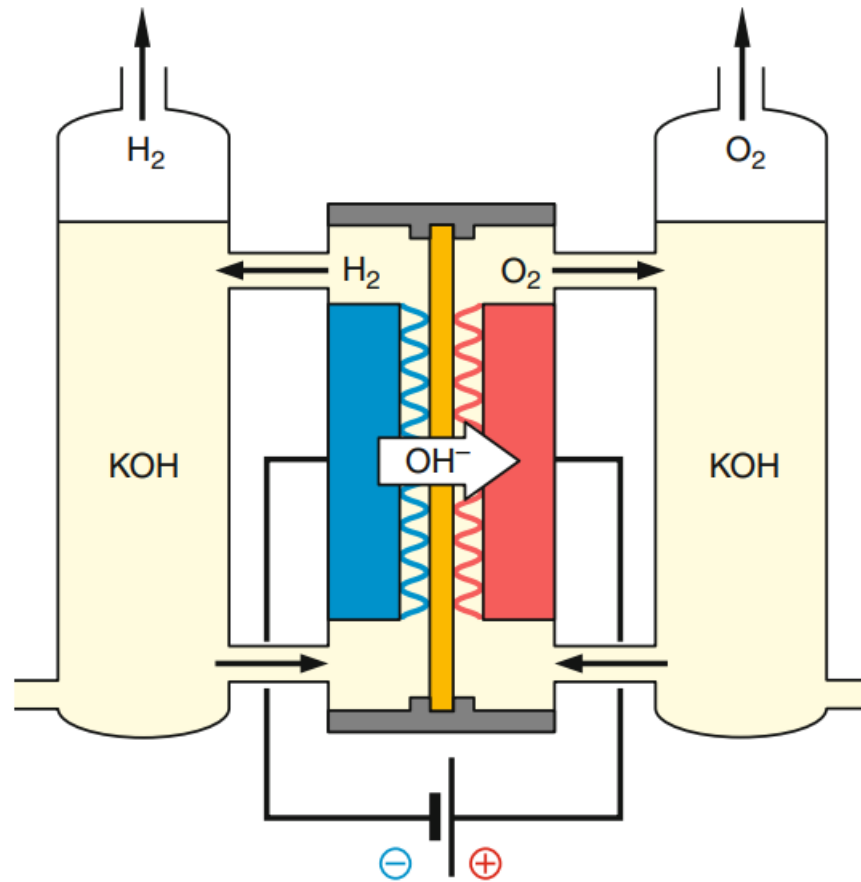
$$E \approx 32,900 \text{ kWh}$$

Endergebnis

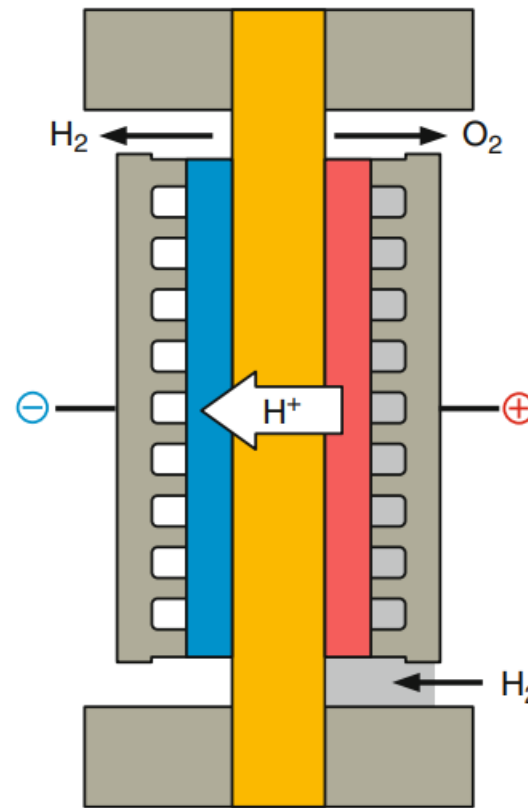
$$E_{\min} \approx 32,9 \text{ MWh/t}_{\text{H}_2}$$

Verschiedene Arten von Hydrolyse

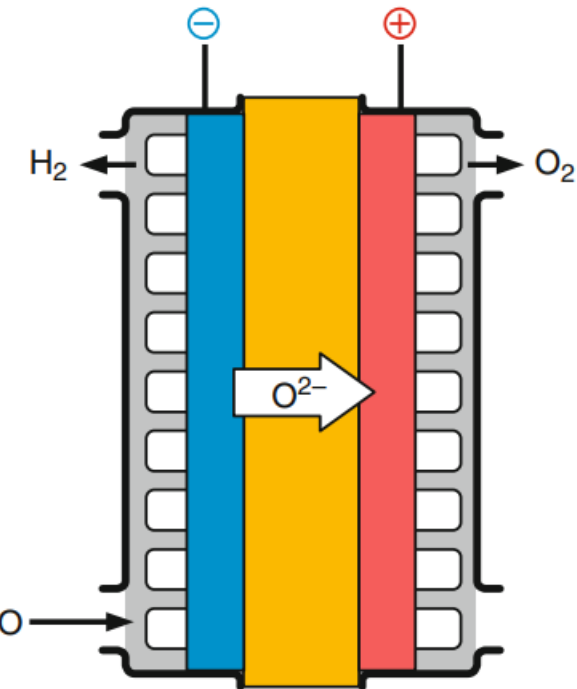
Alkalische Elektrolyse



PEM-Elektrolyse



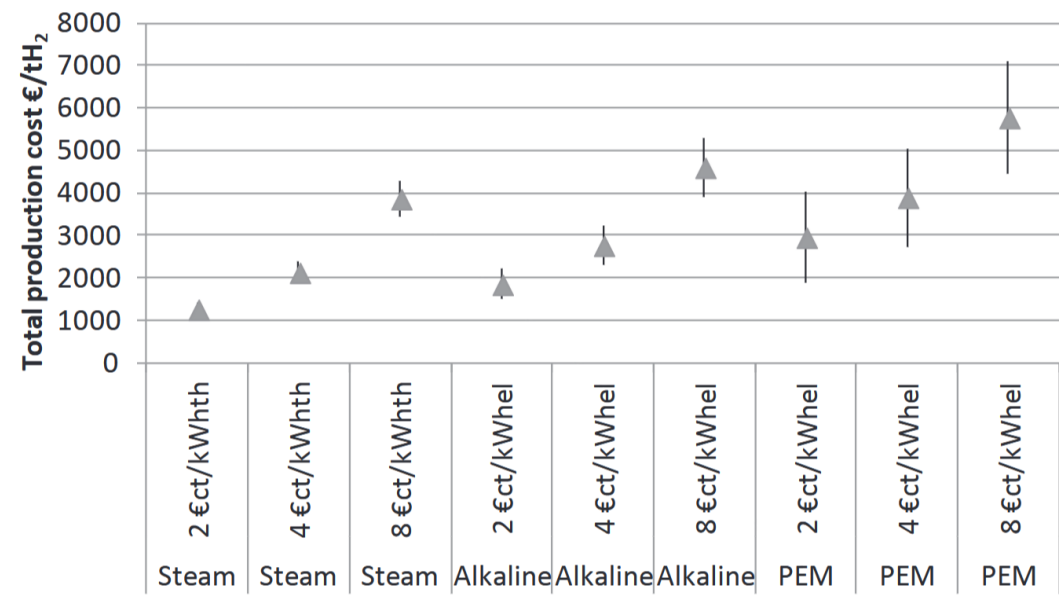
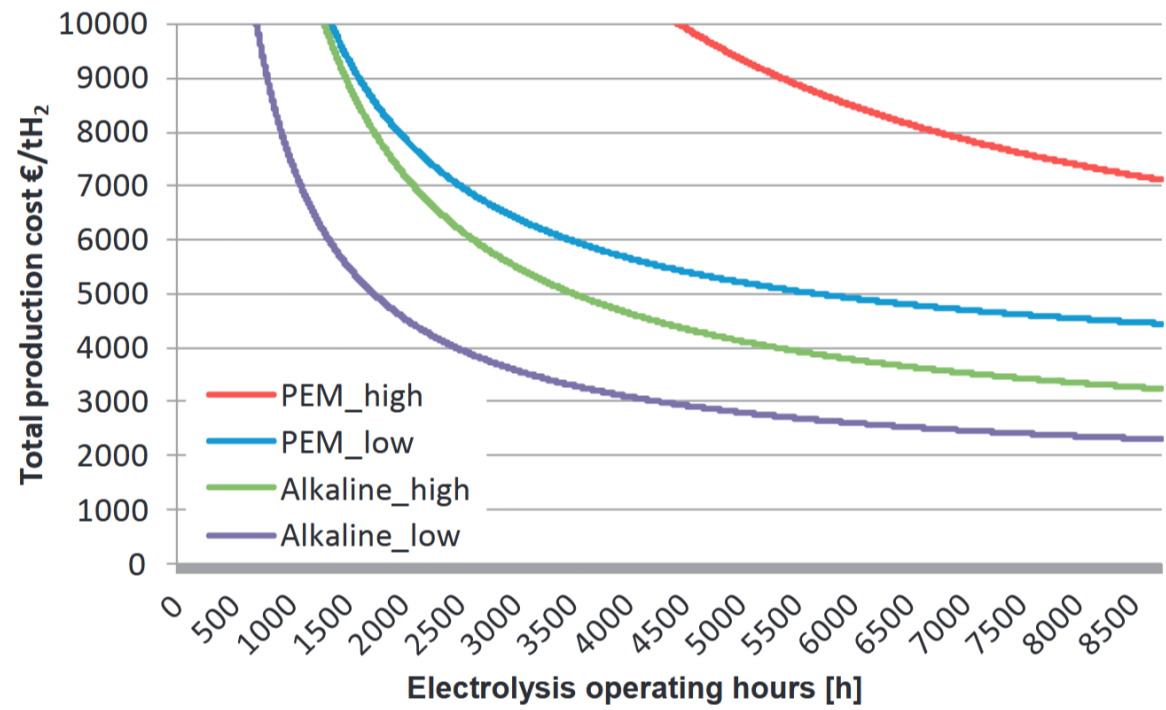
Hochtemperatur-Dampfelektrolyse



Gegenüberstellung

	Alkalie	PEM	SOE
TRL	9	7-8	6-7
Energiebedarf (m³-H₂)	4.3 kWh	4.5-5 kWh	2.6 kWh
Effizienz	65% (LHV)	70% (LHV)	76% (LHV)
Stapelgröße	5.3 MW	1.25 MW	0.1 - 1 MW
Temperatur	40-90°C	20-100°C	700-1000°C
Druck	< 30 bar	<35 bar	Atm.
CAPEX heute [€/kW]	1000-1200	1800-2600	>2000
CAPEX 2050 [€/kW]	370	500	625

Power to X - Ökonomie

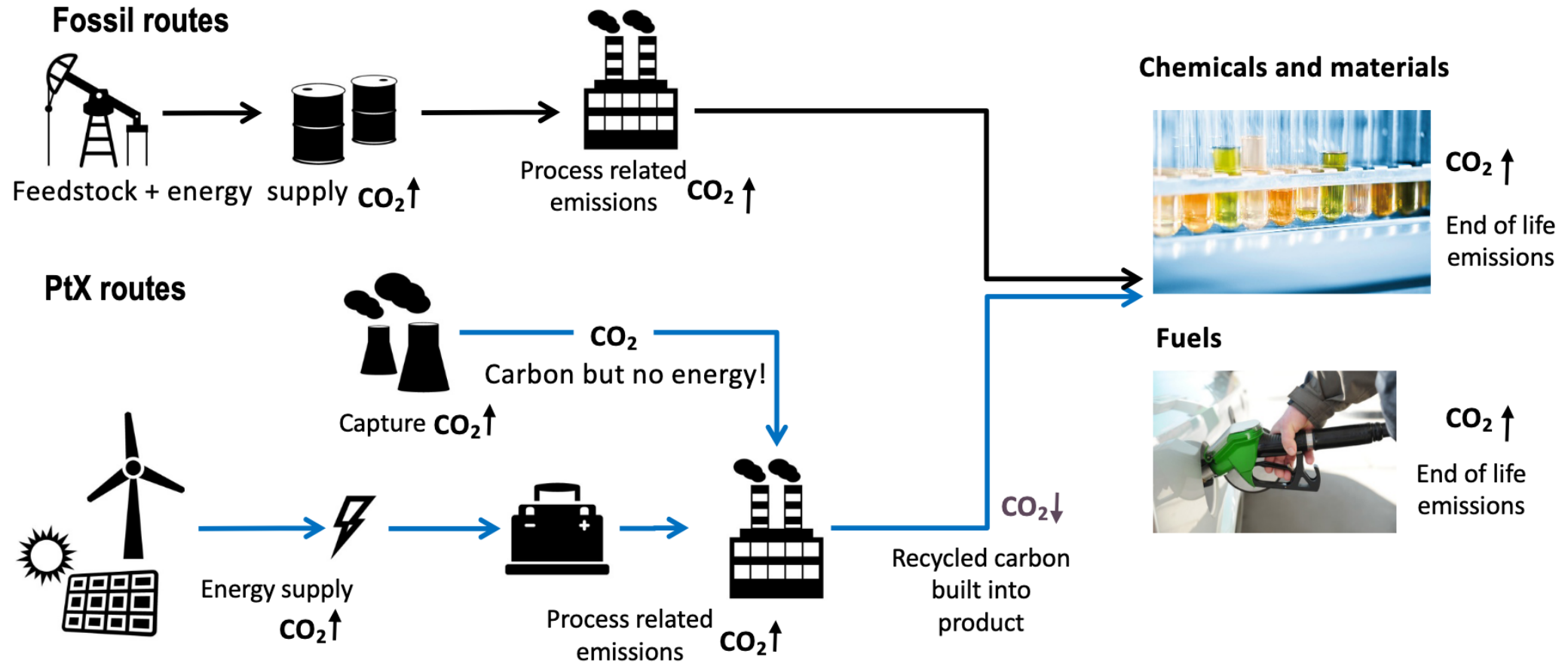




Wasserstoff

E-Fuels

Power to X - Ökonomie



Power to CH₃OH – Carbon Recycling

Mehrere Anlagen

- Island (2012 erste Anlage)
- Norwegen
- China
- Jahresproduktion: 214.000 Tonnen (2023)
- Speziell für Fischerboote
- Jahresproduktion: 400 Millionen Tonnen (2050)



Power to Kerosin



<https://www.atmosfair.de/de/co2-reduzieren/e-kerosin/>

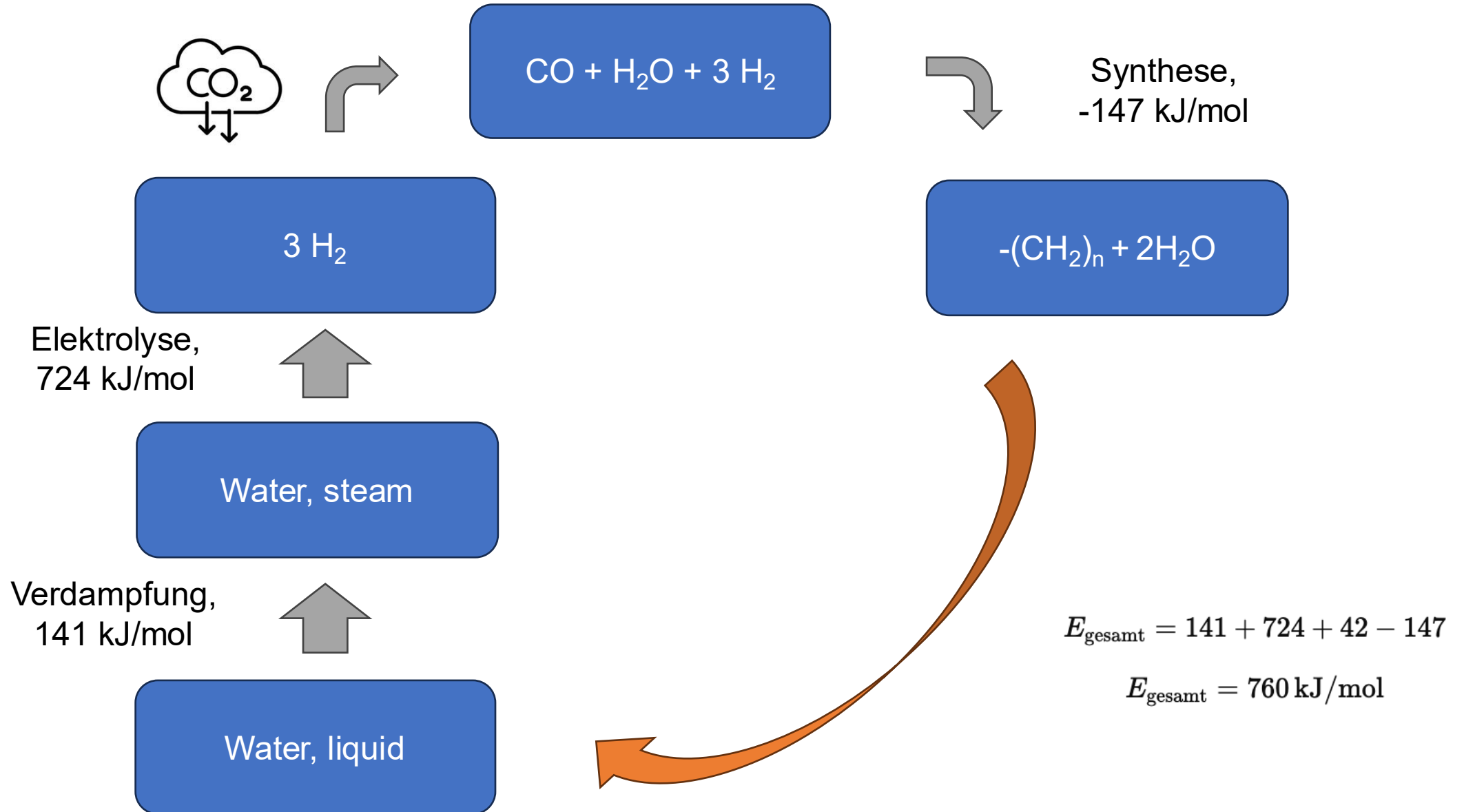


https://ec.europa.eu/assets/cinea/project_fiches/innovation_fund/101132987.pdf

Geplant: 8000 Tonnen

Power to fuels

RWGS,
42 kJ/mol



Power to fuels

$$n = \frac{1,000,000 \text{ g}}{14 \text{ g/mol}}$$

$$n \approx 71,429 \text{ mol}$$

$$E = 71,429 \cdot 760 \text{ kJ}$$

$$E \approx 54,286,000 \text{ kJ}$$

$$E \approx 54,286 \text{ MJ}$$

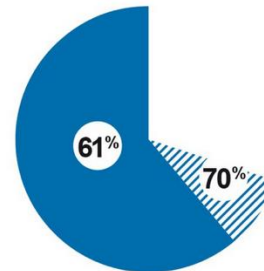
$$E \approx 54,3 \text{ GJ}$$

$$E_{\min} \approx 15,1 \text{ MWh/t Treibstoff}$$

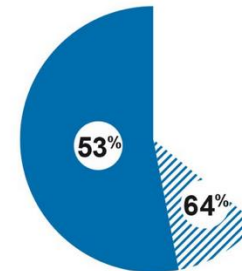
$$\approx 20\text{--}25 \text{ MWh/t}$$

Treibstoff:

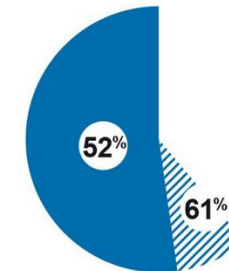
$$\approx 42,000\text{--}43,000 \text{ MJ/t}$$



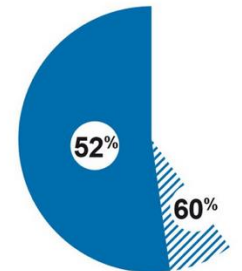
E-Wasserstoff
(gasförmig)



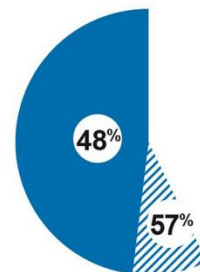
E-Wasserstoff
(verflüssigt)



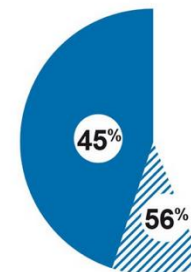
E-Methan
(gasförmig)



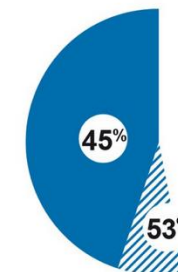
E-Ammoniak



E-Methan
(verflüssigt)



E-Methanol



E-Fuels

Umwandlungseffizienz

Pro eingesetzter Kilowattstunde Strom verbleiben x Prozent im PtX-Produkt





Danke